

BitFlip

CloudSherpa

Software Architecture Specifications

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1 | Introduction

The CloudSherpa Software Architecture Specification elaborates on the architectural and other technical decisions made throughout the development process. Its purpose is to serve as a reference for CloudSherpa's architectural direction. The document is intended for both developers and stakeholders and serves as a mechanism for maintaining alignment between them.

2 | Architectural Requirements

Terminology Note

In the source-code repository, the **Application Service** is implemented by the application named *service*. To distinguish this specific application from the general use of the term service within a service-oriented architecture, it is referred to throughout this document as the Application Service.

2.1 Architectural Patterns

2.1.1 Service Oriented Architecture

CloudSherpa follows the Service Oriented Architecture (SOA) pattern to promote decoupling between the subsystems responsible for cloud data ingestion, the subsystems responsible for processing and facilitating the visualization of this data and the subsystems used to perform intelligence operations. The CloudSherpa architecture identifies three distinct services:

- **Ingestion Service**
The ingestion service is responsible for connecting to the respective cloud provider interfaces for obtaining data from deployments.
- **Application Service**
The Service Service is responsible for the business logic of CloudSherpa, ex-

cutting tasks like data aggregation, authentication and communicating intelligence insights to the Frontend Service.

- **AI/ML Service**

The AI/ML Service is responsible for tasks related to CloudSherpa intelligence, such as Cost and Usage forecasting as well as optimization recommendations.

As part of our architecture we have a Dashboard presentation component that consumes the result of CloudSherpa processing by ways of the Application Service.

2.2 Architectural Diagram

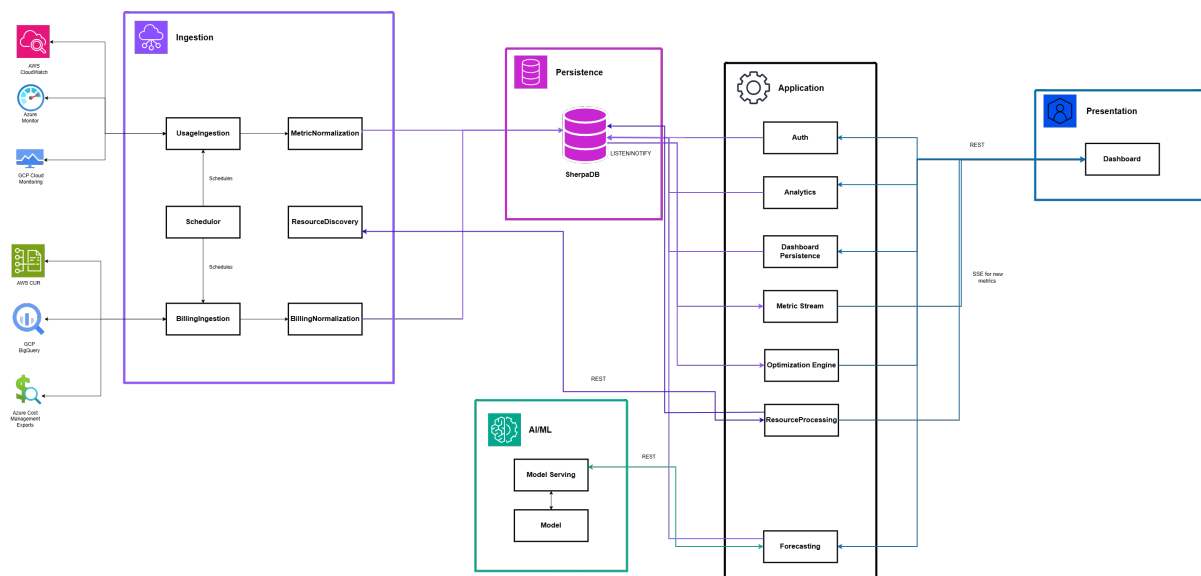


Figure 1: Architectural Diagram

2.3 Entity Relational Diagram

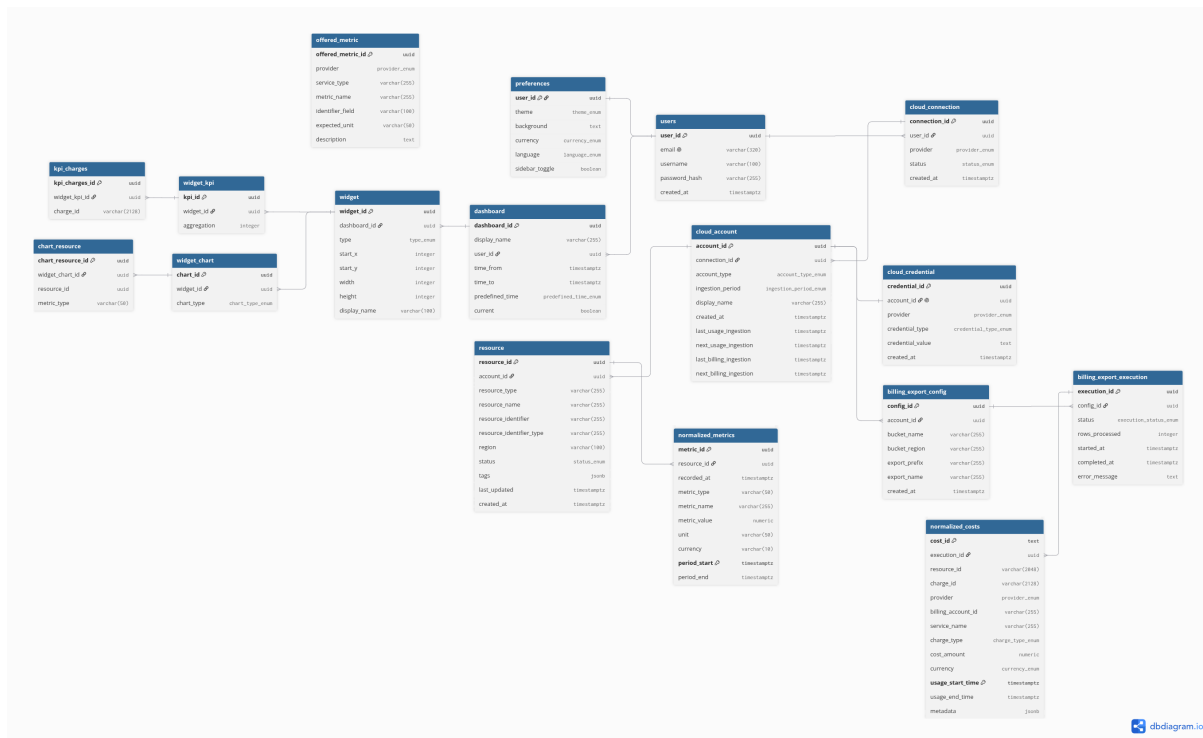


Figure 2: SherpaDB ERD

2.4 Design Patterns

2.4.1 Pipeline Design Pattern for AWS CUR Billing Ingestion

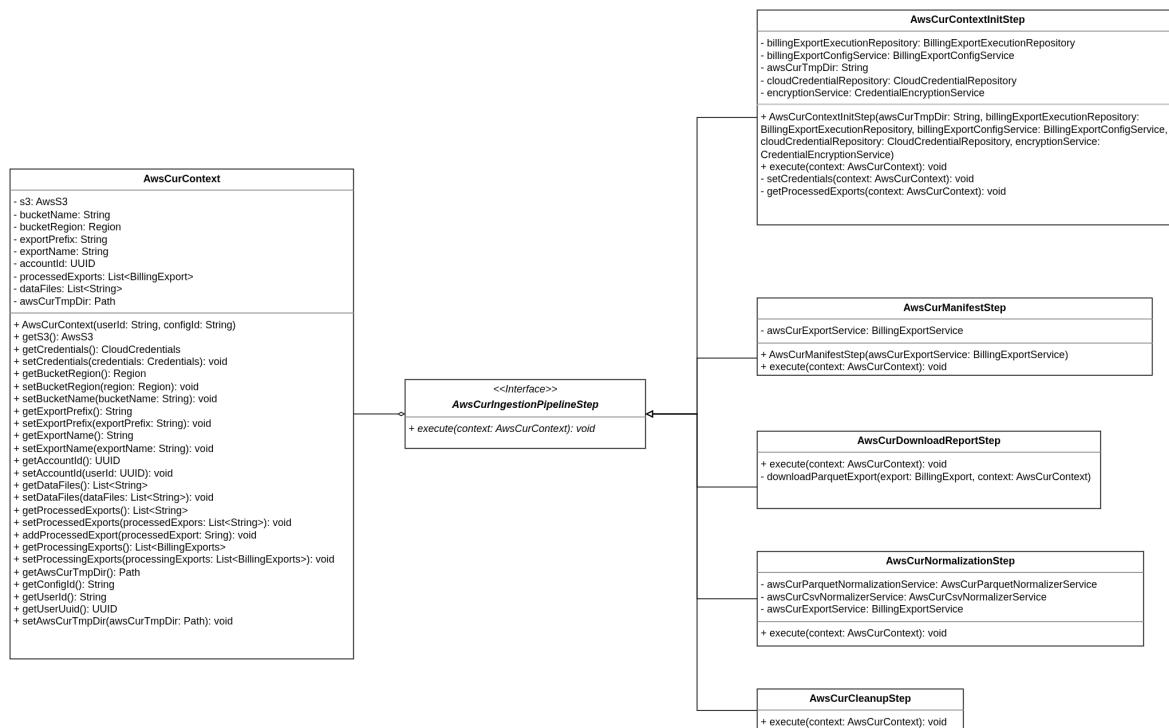


Figure 3: Pipeline Design Pattern For Billing Ingestion

Ingesting reports from a third party, such as a cloud provider, can introduce maintainability challenges. If the provider changes the API used to retrieve a report or modifies the report structure, the ingestion logic must be updated accordingly. Because the nature of future changes cannot be predicted, the ingestion process was implemented using the pipeline design pattern, which divides the process into a sequence of decoupled steps.

The pattern is reflected by the `AwsCurContext` object, which is passed through the pipeline and modified by each implementation of the `AwsCurIngestionPipelineStep` interface. Each step is responsible for a distinct part of the ingestion process.

This design improves maintainability by allowing individual steps to be modified, replaced, or extended without requiring substantial changes to the rest of the ingestion process. The modular structure also makes the billing ingestion process easier to test

and debug.

Note: Not all callers, associations, and aggregations are shown in the diagram.

2.4.2 Builder Design Pattern for Normalized Metrics

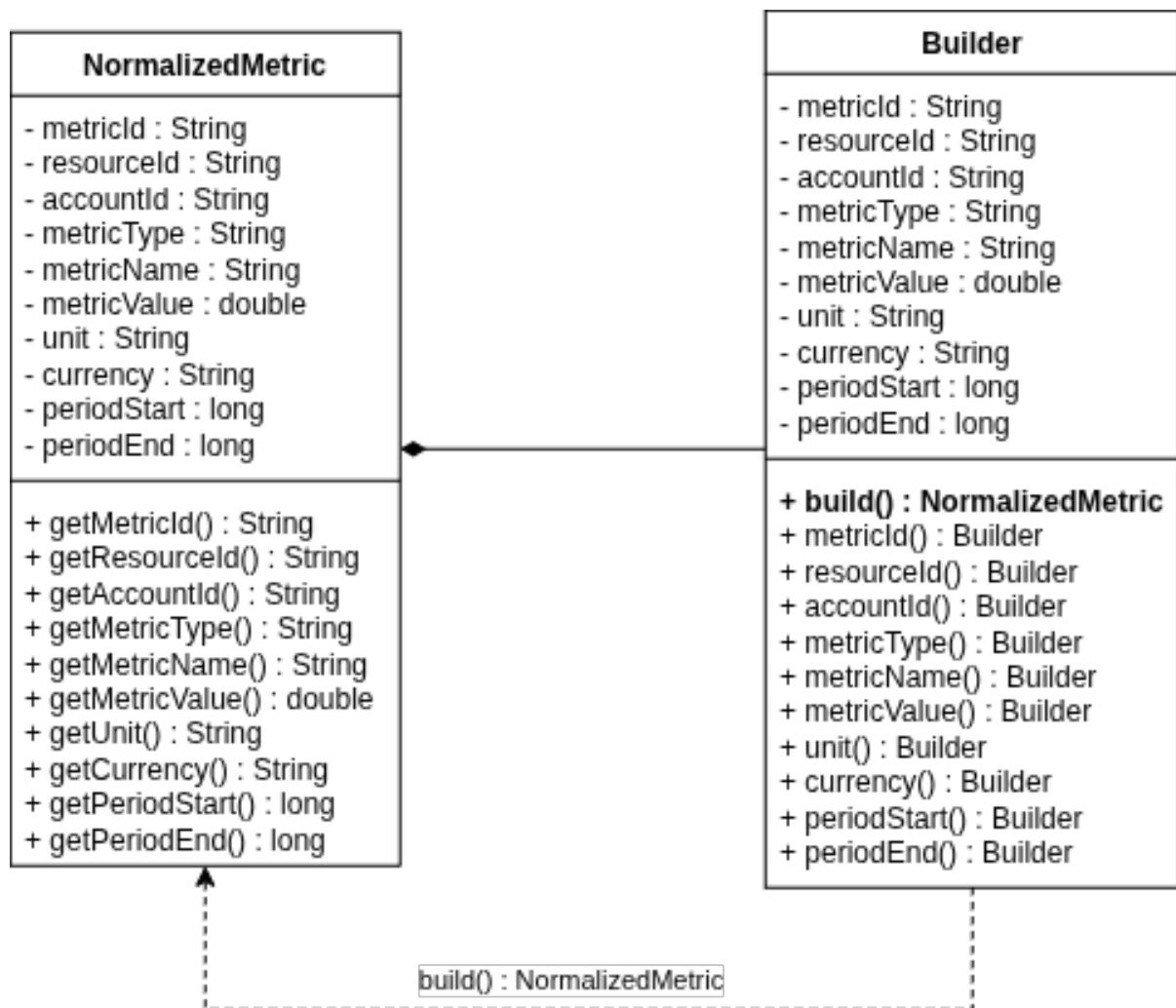


Figure 4: Builder Design Pattern for Normalized Metrics

The `NormalizedMetric` class uses a static nested `Builder` class to create objects using an interface. Each builder method sets a value and returns the builder, allowing methods to be chained together.

The `build()` method creates the final `NormalizedMetric` object using a private constructor. This makes the code easier to read and avoids the need for multiple constructors with different parameter combinations.

2.4.3 Strategy Design Pattern for Resource Discovery

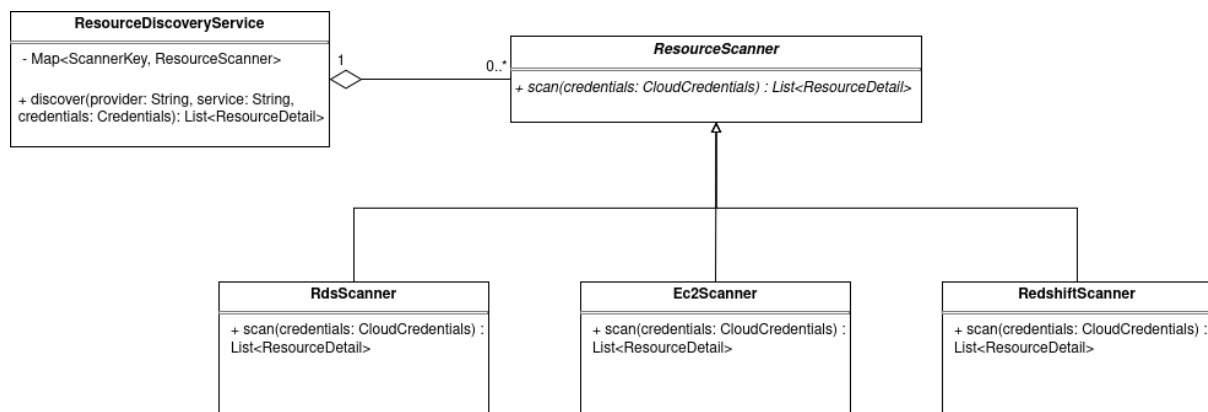


Figure 5: Strategy Design Pattern for Resource Discovery

The `ResourceDiscoveryService` class uses an interface class `ResourceScanner` to treat differing implementations of the `scan` method implemented by the concrete classes such as `Ec2Scanner` uniformly. This allows loose code coupling by avoiding case statements for each type of resource scanner based on the API request parameters.

2.4.4 Registry Design Pattern for scanner selection

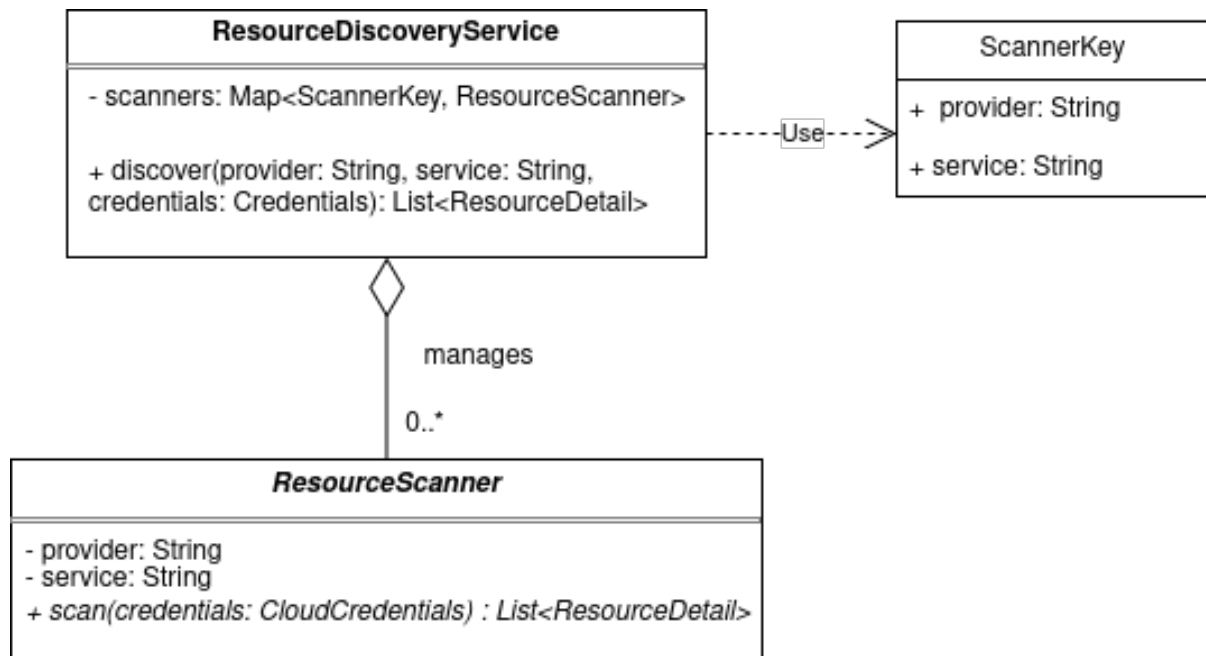


Figure 6: Registry Design Pattern for scanner selection

The `ResourceDiscoveryService` class acts as a registry, containing `<ScannerKey, ResourceScanner>` pairs which allows runtime selection of the Concrete class implementations of `ResourceScanner` interface. This is further enhanced by Springboot which automatically finds `ResourceScanner` concrete classes and adds them to the `ResourceDiscoveryService` scanners, allowing loose code coupling by allowing implementation of new scanners and services without modifying any existing files.

2.5 Constraints

2.5.1 Supported Cloud Providers

The system shall support AWS, Azure and GCP as cloud providers. Cloud providers not listed are outside of the scope of CloudSherpa.

2.5.2 Secure handling of cloud credentials

The system must follow the principle of least privilege when monitoring deployments across cloud providers.

2.5.3 Single Role System (No RBAC)

CloudSherpa does not make provision for role-based users. The developer focus is to be able to deliver a refined system that achieves its primary goal well, which is to serve as a FinOps platform to improve the visibility of and provide insights to cloud spend.

2.5.4 Use Provider APIs Instead of Pushing Data from User Infrastructure

To minimize the operational complexity and security risks associated with collecting usage metrics from CloudSherpa users, cloud provider APIs are used to retrieve data through dedicated IAM or system users, rather than relying on software installed on users' compute instances to push the data.

2.5.5 Cost & Usage Latency

Due to the architectural decision to use cloud provider APIs rather than pushing data from user infrastructure, the frequency and recency of available usage information are subject to provider-specific delays.

A similar constraint applies to billing ingestion. However, unlike usage metrics, billing data cannot be obtained by pushing information directly from user infrastructure.

2.6 Mapping Quality Requirements to Architectural Decisions

Table 1: Mapping Quality Requirements to Architectural Decisions

Quality Requirement (from SRS)	Architectural Decision
NFR1: Performance	
NFR1.1: The system shall retrieve historical metrics for visualization with a p95 request latency of less than 500 ms to ensure a responsive dashboard.	Spread the load of fetching historical metrics by using concurrent reads. Reduce the load of each historical metric series, in particular reducing message overhead by using a downsampling algorithm that reduces the amount of points that need to be included in the message while preserving visual integrity for dashboard visualization.

Quality Requirement (from SRS)	Architectural Decision
NFR1.2: After the initial billing forecast, the system shall retrieve subsequent forecasts within a 30-minute period with a p95 request latency of less than 200 ms to ensure responsive billing forecast pages.	Reuse the expensive forecast operation results by implementing a cache with a TTL of 1 hour. Since billing export ingestion are scheduled for every 6 hours, the potential one hour consistency consequence is accepted as a tradeoff due to how expensive the forecasting operation is.
NFR1.3: The usage ingestion pipeline shall normalize and persist cloud usage metric records at a throughput of at least 1000 records per second under normal operating conditions.	The architecture allows the ingestion service, which is responsible for processing and normalizing cloud billing and usage records, to be scaled independently according to ingestion volume and processing demand, hence improving performance by allocating sufficient resources. The scheduling subsystem allocates background jobs via JobRunr, which manages background asynchronous jobs, allowing multiple usage and ingestion jobs to be processed asynchronously. Our tenant per schema database allows those records to also be written to the database without conflict, due to the fact that separate tables are used for each user's usage and billing metrics. Therefore writes to one users' metric tables do not conflict with writes to other user's metric ingestion.
NFR1.4: The billing ingestion pipeline shall normalize and persist cloud billing records at a throughput of at least 1000 records per second under normal operating conditions.	The architecture allows the ingestion service, which is responsible for processing and normalizing cloud billing and usage records, to be scaled independently according to ingestion volume and processing demand, hence improving performance by scaling up resources. Spread the load by utilizing batch writes and set the tenant context once per multiple writes as opposed to on every write.
NFR2: Scalability	

Quality Requirement (from SRS)	Architectural Decision
NFR2.1: The system should successfully support 100 concurrent users, representing a stress-level load for CloudSherpa, for a period of 1 hour without experiencing more than a 10% degradation in the response times of the below endpoints compared to regular load.	
NFR2.1.1: Historical metric fetch response-time degradation	Data ingestion, processing, and normalization are performed by the independently deployable ingestion service, while dashboard requests are handled by the service layer. The service layer retrieves and aggregates historical data into a form optimized for dashboard rendering (<i>NFR 1.1.</i>), reducing the amount of data transferred and processed for each user request. Because the ingestion and user-facing services are decoupled and operate in isolated execution environments, increases in the number of connected cloud accounts primarily affect the ingestion service rather than the dashboard request path. Each service can therefore be scaled independently according to its workload, helping the system support additional cloud connections and concurrent dashboard users without a significant degradation in response time.
NFR2.1.2: Billing forecasting response-time degradation	Reduce the amount of work completed to limit response-time degradation, the decision made for <i>NFR 1.2</i> applies for <i>NFR 2.1.2</i> as well.
NFR2.1.3: Usage forecasting response-time degradation	Queue incoming forecasting requests to distribute the workload over time and prevent excessive VRAM pressure under concurrent load. (<i>Pending implementation</i>)
NFR3: Security	

Quality Requirement (from SRS)	Architectural Decision
NFR3.1: All sensitive user data and stored cloud provider credentials must be encrypted using TLS 1.3 in transit, and the system must enforce Multi-Factor Authentication (MFA) on 100% of accounts.	All external requests pass through a single controlled entry point before reaching the system's internal services. TLS 1.3 is enforced at this boundary, ensuring that communication between the dashboard and backend services is encrypted in transit.
NFR3.2: The system shall utilize secure, least-privilege identity access management (IAM) strategies, ensuring that CloudSherpa holds read-only access to client AWS, Azure, and GCP billing APIs and ensuring that none of CloudSherpa's requests require write permissions.	Third-party integrations are confined to a single service, allowing them to be implemented through consistent interfaces, governed by provider-specific security configurations, and monitored and audited from a central point. Permissions requests for IAM users include only necessary read-only permissions required for CloudSherpa to successfully discover and monitor selected services as authorized by the user.
NFR3.3: User metrics must remain separated such that no user gains access to resource details or cost and metric information from accounts not belonging to that user, as this information is potentially highly sensitive to individuals or corporations.	The architecture of our database, including our schema-per-tenant tables as well as strict session management and API authorization measures ensure that no user is capable of accessing any tenant specific information, including cost and resource metrics, not belonging to their accounts.
NFR3.4: The system shall have no known high or critical vulnerabilities in third-party dependencies on the main branch.	Automated vulnerability scanners running in the Continuous Integration pipeline
NFR4: Reliability	

Quality Requirement (from SRS)	Architectural Decision
NFR4.1: In the event of a third-party cloud provider API failure or timeout (e.g., an AWS CloudWatch outage), the ingestion pipeline shall automatically retry the connection and keep track of the failure period, ensuring 0% duplication or metric loss once the provider is back online.	The CloudSherpa ingestion scheduler uses a recurring database poll to find accounts due for ingestion, using database fields to keep track of the last successful ingestion as well as the next scheduled events. JobRunr background jobs are configured to retry requests three times upon failure, and database success fields are only updated upon success, ensuring that upon subsequent scans outstanding jobs will be handled when the external provider is back online.
NFR5: Maintainability	
NFR5.1: The codebase must maintain a minimum of 80% automated test coverage across all core components.	The modularity of our architecture improves the ease and pace at which tests can be developed, as large parts of our system remain loosely coupled. Separate CI workflows ensure that tests can be run for subsystems selectively.
NFR5.2: Static Analysis Should show zero maintainability issues for CloudSherpa services	Well defined domain boundaries ensure that code smells are isolated to modular components.
NFR6: Usability	
NFR6.2: The CloudSherpa dashboard shall render correctly on modern web browsers to ensure plain-language summaries remain highly accessible.	CloudSherpa was built using modern standards-compliant web technologies using features that are supported by all modern web browsers. Responsive CSS and the use of high quality component libraries such as Apache ECharts and Shadcn ensure that components render correctly on differing screen sizes and resolutions, and lighthouse reports are routinely run to ensure performance and accessibility. Playwright end-to-end tests verify that the dashboards functions on multiple web browsers.
NFR7: Availability	

Quality Requirement (from SRS)	Architectural Decision
NFR7.1: The dashboard shall maintain an availability of at least 95% during the defined availability period.	CloudSherpa's services are separately deployable, with the database being the primary single point of failure. If one service degrades or becomes unavailable, the remaining services can continue operating independently. The health of the services are continuously monitored during the availability periods.
NFR7.2: The application service, responsible for all functionality excluding ingestion and forecasting, shall maintain an availability of at least 95% during the defined availability period.	CloudSherpa's services are separately deployable, with the database being the primary single point of failure. If one service degrades or becomes unavailable, the remaining services can continue operating independently. The health of the services are continuously monitored during the availability periods.
NFR7.3: The ingestion service, responsible for ingesting cloud usage and billing data, shall maintain an availability of at least 95% during the defined availability period.	CloudSherpa's services are separately deployable, with the database being the primary single point of failure. If one service degrades or becomes unavailable, the remaining services can continue operating independently. The health of the services are continuously monitored during the availability periods.
NFR7.4: The intelligence service, responsible for forecasting functionality, shall maintain an availability of at least 95% during the defined availability period.	CloudSherpa's services are separately deployable, with the database being the primary single point of failure, however the intelligence service remains unaffected during database degradation. If one service degrades or becomes unavailable, the remaining services can continue operating independently.

2.7 NFR Traceability Matrix

⚠ NFR Testing Scope

Due to time constraints, not all documented non-functional requirements have been fully verified through executable tests for Demo 3. Outstanding NFR tests are planned for completion before the next demo.

Table 2: NFR Traceability Matrix

ID	Quantified Requirement	Tactic in SAS	Test / Tool	Target / Actual
NFR1.1	Historical metric retrieval p95 latency at dashboard load.	Reduce load with efficient processing via LTTB downsampling	Grafana k6	< 500ms / 272.2ms
NFR1.2	Billing forecast p95 latency for subsequent requests within 30 minutes.	Aggressive cache with TTL 30 minutes	Grafana k6	< 200ms / 6.9ms
NFR1.3	Usage ingestion throughput under normal load.	Asynchronous ingestion	Spring Boot Integration Test	>= 1000 records/s / TBD
NFR1.4	Billing ingestion throughput under normal load	Efficient processing via batching	Spring Boot Integration Test	>= 1000 records/s / GCP: 955 records/s, AWS: TBD, Azure: TBD,
NFR2.1: 100 concurrent users for 5 minutes with $\leq 10\%$ response-time degradation				
NFR2.1.1	Historical metric fetch response-time degradation	Reduce load with efficient processing via LTTB downsampling	Grafana k6	$\leq 10\%$ / 3300.70%
NFR2.1.2	Billing forecasting response-time degradation	Aggressive cache with TTL 30 minutes	Grafana k6	$\leq 10\%$ / TBD
NFR2.1.3	Usage forecasting response-time degradation	No particular tactic, validated by test	Grafana k6	$\leq 10\%$ / TBD

ID	Quantified Requirement	Tactic in SAS	Test / Tool	Target / Actual
NFR3.1	TLS 1.3 enforced at external ingress.	Encrypted client-to-ingress communication	openssl connect	TLS 1.3 / TLS 1.3, TLS 1.2
NFR3.2	Read-only cloud provider access	Least-privilege IAM policies	Read-only subsystems deny write	write action denied / TBD
NFR3.3	Cross-tenant data isolation	Schema-per-tenant isolation with authenticated access control	Spring Boot Integration Test	pass / TBD
NFR3.4	High/critical dependency vulnerabilities	Automated dependency scanning and updates	GitHub dependabot, NPM audit, OWASP maven dependency check	0 reported high or critical vulnerabilities / dependabot: 0, npm audit: 0, dependency check: TBD
NFR4.1	Cloud provider API failure recovery without metric loss or duplication.	Retry failed interval with backoff after three failed attempts. Continuous retry and interval extension to cover failure period whilst limiting external API hits	Spring Boot Test	pass / pass
NFR5.1	Automated test coverage across core components	Automated testing enforced in CI	SonarQube & coverage reports	≥ 80% / Application: 45.2%, Ingestion: 67.9%, Intelligence: 73.70%
NFR5.2	Static-analysis maintainability issues	Automated static analysis in CI	SonarQube	0 issues / Ingestion: 0, Application: 0, Intelligence: 0, Dashboard: 43

ID	Quantified Requirement	Tactic in SAS	Test / Tool	Target / Actual
NFR6.2	Cross-browser dashboard compatibility	Standards-compliant responsive web interface	Playwright E2E tests & Supported browsers pass	pass / pass
NFR7: Minimum 95% availability during the defined availability period				
NFR7.1	Dashboard availability	Independent service deployment and health monitoring	Uptime Kuma	≥ 95% / 81.37%
NFR7.2	Application service availability	Independent service deployment and health monitoring	Uptime Kuma	≥ 95% / 81.39%
NFR7.3	Ingestion service availability	Independent service deployment and health monitoring	Uptime Kuma	≥ 95% / 81.35%
NFR7.4	Intelligence service availability	Independent service deployment and health monitoring	Uptime Kuma	≥ 95% / 99.99%

3 | Technology Requirements

Terminology Note

Please note that references such as **FR X** or **NFR X** refer to requirements documented in the CloudSherpa Software Requirement Specification documentation.

3.1 Next.js

We chose to use the Next.js full-stack framework for the CloudSherpa user-facing **Dashboard**.

3.1.1 Supporting functional requirements

- **FR 5. Web-Based Dashboard**

Next.js is a framework intended for web development and thus facilitates the development of our web-based Dashboard as elicited by **FR 5** in our SRS document. Reactive dashboard components are enabled by Next.js since it is built on top of the React framework.

- **FR 6. User and Account Management** Next.js middleware can inspect incoming HTTP requests and redirect users to the login page when an authorization token is absent.
- **FR 7. Alerts and Notifications** Next.js integrates with shadcn/ui, which provides reusable and accessible interface components for displaying alerts and notifications to users.

3.1.2 Supporting non-functional requirements

- **NFR 1. Performance**

Next.js Server Components can be leveraged for computationally intensive Dashboard tasks, limiting client-side computation and improving dashboard responsiveness.

- **NFR 3. Security**

Next.js Server Components allows developers to omit potentially sensitive information from the bundle that gets sent to the client browser. Next.js middleware allows code to be executed server-side before a request is completed, allowing developers to intercept and inspect incoming client requests and perform custom actions such as redirecting the user to the login page when the authorization token is omitted from the request.

- **NFR 6. Usability** Next.js integrates with shadcn/ui, our chosen component library, which provides reusable and accessible interface components that support the development of a consistent and user-friendly dashboard.

3.1.3 Ease of development

- **App Router**

The Next.js App Router simplifies navigation development by using the application's directory structure to define routes. It also provides conventions, such as route groups and private folders, that allow developers to organize related files and routes without necessarily affecting the resulting URL structure.

3.2 Spring Boot

We selected Spring Boot with Java 21 for the CloudSherpa Ingestion and Application services because of its performance benefits compared with the considered alternatives, such as NestJS, as well as Java's strong support for data structures and algorithms and the SDKs provided by the cloud providers that CloudsSherpa supports.

3.2.1 Supporting functional requirements

- **FR 1. Cloud Data Integration**

AWS, GCP, and Azure provide Java SDKs that CloudSherpa uses for resource discovery and the ingestion of usage metrics and billing data. The JobRunr Java library provides persistent, multithreaded background job processing, which we use to schedule and execute usage and billing ingestion tasks.

- **FR 2. Data storage and processing**

Spring Boot provides object-relational mapping support through JPA and Hibernate, allowing database records to be represented and accessed as Java objects. Hibernate also supports schema-based multitenancy, enabling tenant data to be isolated in separate database schemas. It also integrates with PostgreSQL's LISTEN/NOTIFY functionality, enabling event-driven processing in response to database changes.

- **FR 6. User and Account Management**

Utilizing Spring Security Filter Chain for managing authorization, users and user-related context can be obtained for each request as part of the Authentication Context facilitated by Spring Security and the JWT protocol.

3.2.2 Supporting non-functional requirements

- **NFR 1. Performance**

The spring boot multi-threaded architecture allows for competitive performance.

- **NFR 2. Scalability** Spring Batch supports the efficient processing of large data volumes through chunk-based processing and parallel execution.

- **NFR 3. Security**

Spring Security provides a pipe-and-filter architecture to ensure that unauthenticated or unauthorized requests never reach the controllers but gets filtered out at an earlier stage in the Spring Security Filter Chain.

- **NFR 4. Reliability**

Spring Boot Actuator exposes health and additional monitoring endpoints out of

the box, enabling health-checks and uptime monitoring. Java custom exception hierarchies allows developers handle exceptions gracefully.

- **NFR 5. Maintainability**

OOP and Java packages enables modular development making code easier to test, maintain and extend. Supporting testing frameworks such as *JUnit*, *Mockito* and *AssertJ* enable unit and integration testing.

3.2.3 Ease of development

- **Familiarity**

As team BitFlip is comprised of 5 university students with a lot of exposure to the Java programming language, we are apply to better apply learnt concepts during the development of the most technical parts of CloudSherpa.

3.3 Docker and Docker Compose

3.3.1 Supporting non-functional requirements

- **NFR 1. Performance**

Docker shares the operating system kernel which results in better performance than using traditional VMs.

- **NFR 2. Scalability**

Containers can easily be scaled up or down. Containers can be replicated to facilitate distributed runtime environments.

- **NFR 3. Security**

Our application runs as a set of isolated containers. Isolated containers have the benefit of increased security: when one part of the system is compromised it does not automatically mean that the entire system is compromised. Taking this into account, we explicitly handle all sensitive operations within a part of the system that runs in an isolated container and is less likely to be vulnerable to malicious exploits.

3.3.2 Ease of development

- **Portability**

Docker was chosen to improve application portability by reducing its dependence on the environment in which it is executed.

- **Reproducibility**

Containers isolate the application and its dependencies from the host environ-

ment. As a result, container behaviour should remain consistent across different environments, enabling more predictable and reproducible application execution.

3.4 PostgreSQL & TimescaleDB

We use **PostgreSQL** as our relational database management system (RDBMS), together with the TimescaleDB extension for storing and querying time-series data. A key reason for selecting PostgreSQL is that TimescaleDB is built on top of it, allowing time-series hypertables and conventional relational tables to coexist and be queried within a single database.

3.4.1 Supporting functional requirements

- **FR 2. Data Storage and Processing**

PostgreSQL schemas define the structure and organization of data stored at rest. The TimescaleDB extension is used to efficiently store and query time-series data, while PostgreSQL's `LISTEN/NOTIFY` functionality supports event-driven communication when new usage metric data becomes available. A schema-per-tenant architecture is used to logically isolate each tenant's data within the database.

3.4.2 Supporting non-functional requirements

- **NFR 1. Performance**

TimescaleDB improves the storage and query performance of time-series data through time-based partitioning and optimizations designed for time-series workloads.

3.5 FastAPI

FastAPI was chosen to facilitate communication between the forecasting model and the application service. Its Python runtime enables direct integration with the forecasting model (Chronos-2) and its supporting machine-learning libraries, while its performance and ease of development support the efficient implementation of the required API.

3.5.1 Supporting Functional Requirements

- **FR 3. Cost Prediction**

The FastAPI framework facilitates REST-based communication between the Application Service and the AI/ML Intelligence Service. The Application Service sends time-series data and a specified prediction horizon to the forecasting model

exposed through FastAPI, which returns predictions for the requested number of future timestamps.

3.5.2 Supporting Non-Functional Requirements

- **NFR 1. Performance** According to the official FastAPI documentation, its performance is comparable to that of high-performance frameworks built using technologies such as Node.js and Go. As FastAPI is primarily responsible for exposing the forecasting model through a REST API, its performance helps minimize the communication overhead between the Application Service and the AI/ML Intelligence Service.

3.5.3 Ease of Development

Built on Pydantic, FastAPI automates development tasks such as validation, serialization and endpoint documentation.

4 | Deployment

4.1 Deployment Requirements

- **Live, Accessible System:**
The system is accessible at <https://cloudsherpa.gjjcs.org>
- **Environment Parity:**
 - **Development Environment**
BitFlip members develop CloudSherpa on their local machines. To ensure reproducibility and consistency across the team, the development environment is containerized using Docker and Docker Compose. This setup closely mirrors the production deployment environment, particularly in how container images are built and services are orchestrated. Team members who do not have sufficient local computing resources to run the complete containerized environment can instead make use of the development servers provided by the frameworks that we use (refer to section 3).
 - **Staging Environment**
The CloudSherpa staging environment is hosted on an AWS EC2 instance, with deployment automated through GitHub Actions. It serves as a Quality

Assurance (QA) environment in which the team can test the platform under conditions that closely resemble the production environment. Nginx-enforced HTTP Basic Authentication restricts access to authorized team members and stakeholders, preventing unintended public access to the staging deployment. The automated deployment workflow is triggered by a push event to the `staging` branch.

- **Production Environment**

The CloudSherpa production environment is hosted on an AWS EC2 instance, with deployment automated through GitHub Actions. The production deployment workflow is triggered by a push event to the `main` branch.

- **Containerisation:**

Docker was selected as the containerisation platform for CloudSherpa, enabling the system to be developed and deployed as a set of isolated, portable services. Docker Compose is used to define and manage the platform's multi-container architecture. For deployment, AWS Elastic Container Registry (ECR) serves as a private, managed container registry that integrates seamlessly with the staging and production environments hosted on AWS.

- **Secrets Management:**

Local `.env` files are used to manage secrets and environment-specific configuration. To ensure consistency across environments and enable configuration validation, the project uses environment variable schemas. *Varlock* was selected to define and manage these schemas, together with a custom validation script that verifies that all required environment variables are present. The Varlock schema files can be safely committed to version control because they define only the expected configuration structure and do not contain any secret values.

- **Rollback Strategy:**

CloudSherpa container images are built and deployed automatically using GitHub Actions. Each image is tagged with the commit hash associated with the deployment, allowing deployments to be pinned to specific versions of the codebase. A deployment script accepts a commit hash as a command-line argument, pulls the corresponding images from the container registry, and deploys them. If a rollback is required, the commit hash of a previously stable deployment is supplied to the script, restoring the system to that version.

4.2 Deployment Diagram

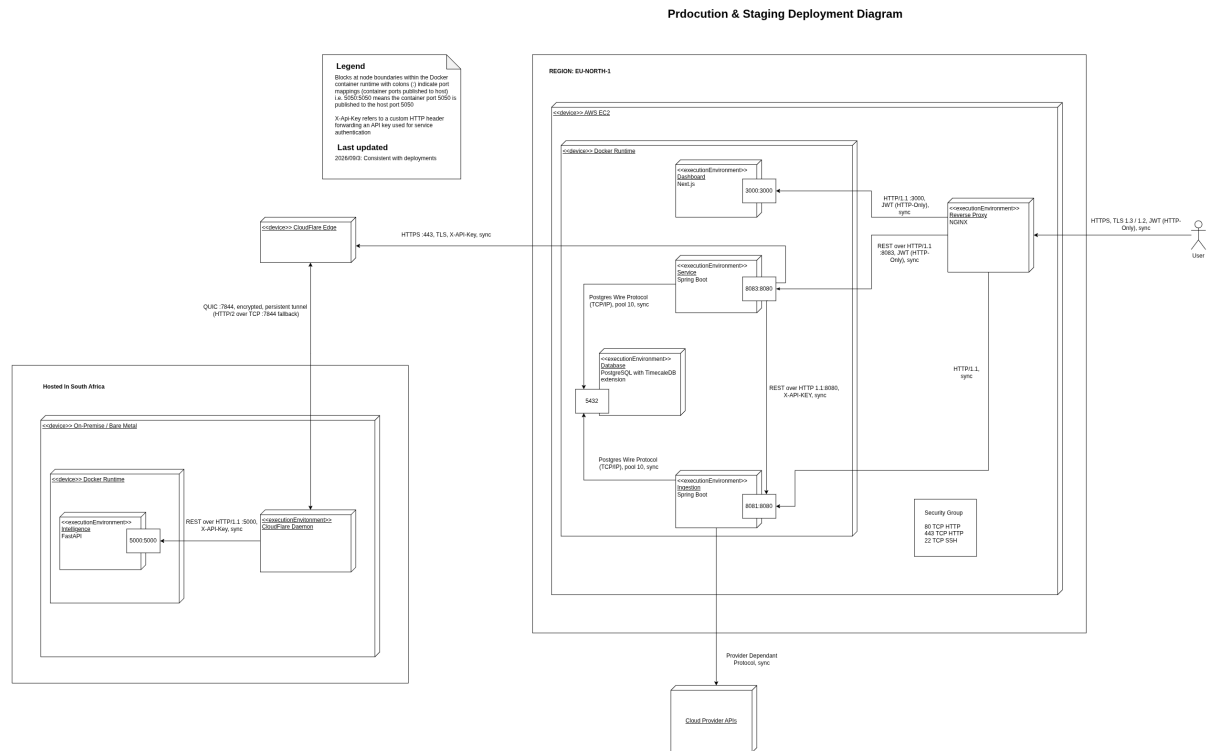


Figure 7: CloudSherpa Deployment Diagram (Production)

Note: Please zoom in when viewing the PDF to inspect diagram details.

4.3 CI/CD Pipeline Diagram

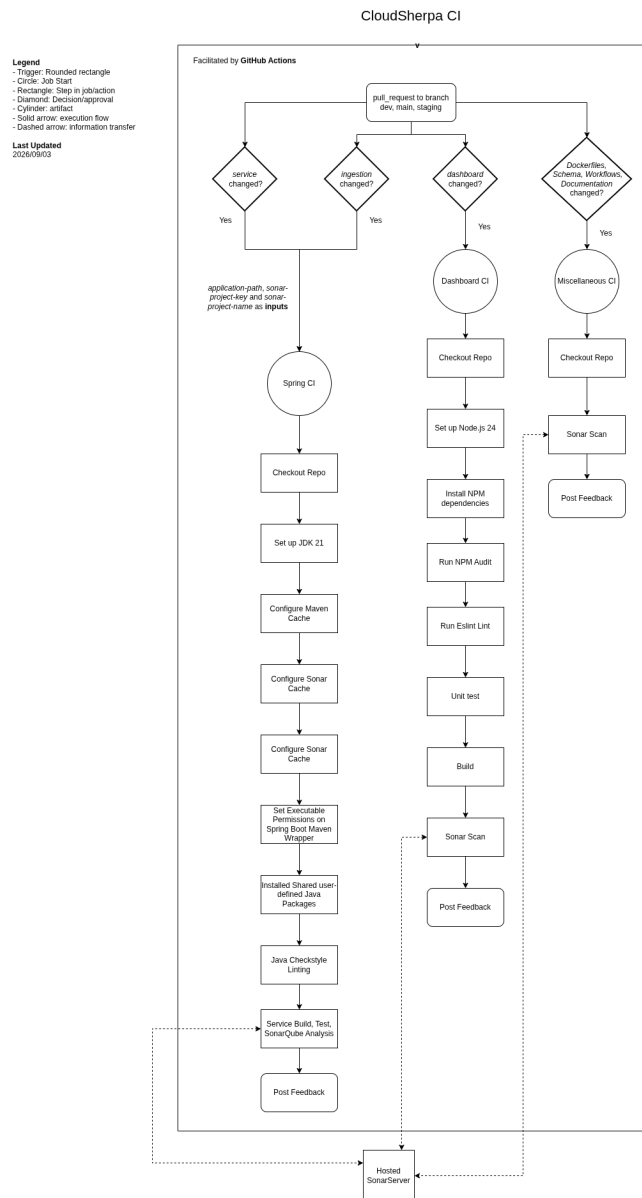


Figure 8: CI Pipeline Flow

CloudSherpa CD

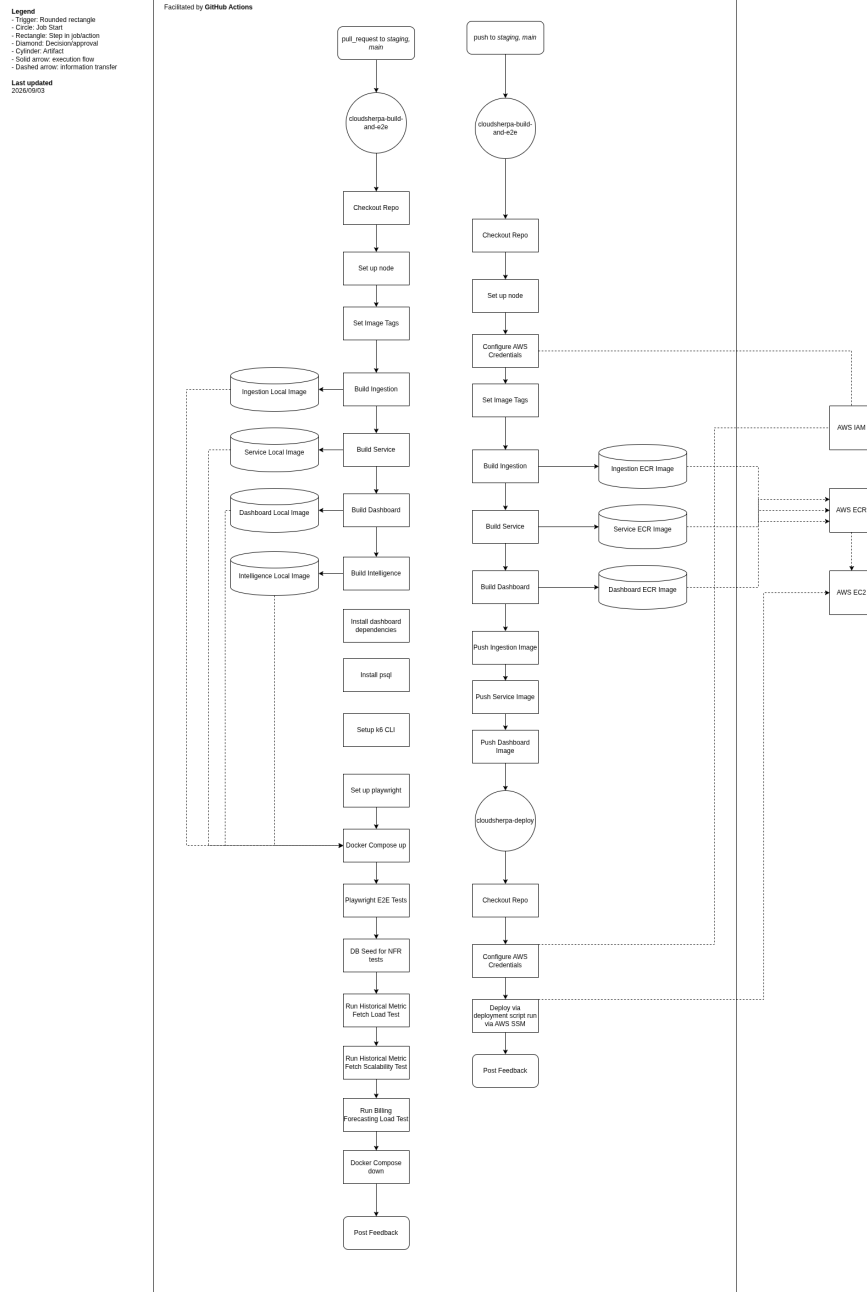


Figure 9: CD Pipeline Flow

Note: Please zoom in when viewing the PDF to inspect diagram details.

A | Appendix A: Architecture Decision Records

A.1 ADR - 001

ADR - 001

Title: Ingestion Scheduler job management

Status: Accepted

Context: Ingestion jobs from multiple cloud accounts must be ingested at varying times dependent on user-selected intervals. These events must not interfere with the rest of the subsystem's functioning. Failure conditions must be handled to ensure resiliency and avoid data loss in the case of external outages.

Alternatives:

1. Use Postgress listen/notify to create one thread per due account (could lead to thread explosion and deployment costs, no outage protection, satisfies independent functioning, tightly coupled to database).
2. Use database polling with timestamped fields and a thread pool to queue due account jobs. (No thread explosion, outage protection, no retry logic, manual retry logic and exception handling required).
3. Use database polling with timestamped fields and a background job manager (JobRunr) to manage threads. (No thread explosion, outage protection, automatic retry attempts, detailed logs maintained, additional storage required, dependency)

Decision: Database polling with timestamped fields and JobRunr background job manager

Consequence:

- + Threads are managed by JobRunr to avoid thread explosion.
- + JobRunr manages retry logic automatically for resiliency.
- + Detailed logging is available through Jobrunr generated tables.
- + Polling allows jobs to be handled multiple times in case of failure
 - JobRunr handles retries uniformly, potentially limiting future custom handling
 - JobRunr adds overhead and logs may incur additional storage costs

A.2 ADR - 002



ADR - 002

Title: PostgreSQL LISTEN/NOTIFY for inter-service communication

Status: Accepted

Context: The ingestion subsystem must notify the service subsystem when metrics have been added. The communication mechanism should minimise deployment complexity while remaining sufficiently reliable for the expected workload.

Alternatives:

1. Use an event broker (e.g. Kafka or RabbitMQ). (Mature event-driven communication, durable messaging, scalable, increased operational complexity and additional infrastructure).
2. Use PostgreSQL LISTEN/NOTIFY. (Simple deployment, no additional infrastructure, lightweight, notifications are transient and less suitable for large-scale event streaming).

Decision: Use PostgreSQL LISTEN/NOTIFY for communication between the ingestion and service subsystems.

Consequence:

- + Reduces operational complexity by avoiding additional messaging infrastructure.
- + Reduces deployment size and simplifies maintenance.
- + Integrates naturally with the existing PostgreSQL database.
- Notifications are not durable and may be missed if consumers are unavailable.
- Less suitable than dedicated message brokers for future high-volume event streaming.

A.3 ADR - 003



ADR - 003

Title: Server-Sent Events for real-time dashboard updates

Status: Accepted

Context: Users should metric updates without manually refreshing the dashboard or generating unnecessary requests.

Alternatives:

1. Poll the server periodically. (Simple implementation, wasted requests when no updates exist, increased server load).
2. Use Server-Sent Events (SSE). (Real-time updates over a persistent HTTP connection, lower network overhead, persistent connections must be maintained).

Decision: Use Server-Sent Events (SSE) to push ingestion status updates to the dashboard.

Consequence:

- + Eliminates unnecessary polling requests.
- + Provides near real-time updates to users.
- Persistent HTTP connections consume server resources.
- Connections must be re-established after network interruptions.

A.4 ADR - 004



ADR - 004

Title: Cloud billing exports for cost analysis

Status: Accepted

Context: Cloud resource costs can either be approximated from usage metrics or obtained directly from provider billing exports. Accurate historical costs are required for reporting.

Alternatives:

1. Approximate costs using collected usage metrics. (Near real-time estimates, numerous unknown pricing variables reduce accuracy, metric specific logic implementation).
2. Use cloud provider billing exports. (Accurate provider-calculated costs, delayed availability due to billing export schedules).

Decision: Use provider billing exports as the source of cloud cost information.

Consequence:

- + Provides accurate provider-calculated billing values.
- + Avoids implementing and maintaining complex pricing logic.
- + Produces consistent historical reporting.
- Users accept latency before costs become visible.

A.5 ADR - 005



ADR - 005

Title: Use of foundational prediction models

Status: Accepted

Context: CloudSherpa provides projections of future metric values of cloud resource utilisation based on historical data. The prediction model could either be developed and trained specifically for the system or utilise an existing foundational model.

Alternatives:

1. Develop a custom prediction model. (Highly sophisticated models could achieve higher quality predictions if provided sufficient data, significant development effort, extensive training data required, skill requirement with parameter tuning or method selection).
2. Use an existing foundational language model. (High-quality predictions based on large datasets, rapid development, dependency on third-party model providers).

Decision: Use a foundational prediction model to predict future metric values based on historical data.

Consequence:

- + Significantly reduces development effort.
- + Produces high-quality predictions based on extensive training set.
- Creates dependency on external model providers.
- Less tuning and optimization according to specific customer profiles possible.

A.6 ADR - 006

ADR - 006

Title: Adopt a schema-per-tenant multi-tenancy architecture

Status: Accepted

Context: CloudSherpa must support a growing number of tenants while maintaining logical data isolation, efficient query performance, and manageable operational overhead. The chosen multi-tenancy strategy should balance scalability, maintainability, and performance as the number of tenants and stored historical cloud metrics increases.

Alternatives:

1. Database-per-tenant. (Strong physical isolation and independent administration, however managing potentially thousands of databases significantly increases operational complexity, resource consumption, backups, monitoring, and schema migrations.)
2. Shared schema with tenant identifiers. (Simplified database administration with a single schema, however all tenant data resides in the same tables, resulting in larger indexes, tenant filtering on every query, increased risk of accidental cross-tenant data access, and reduced query locality.)
3. Schema-per-tenant. (Provides logical isolation while sharing the same database instance, resulting in smaller tenant-specific tables, improved query performance, and significantly lower administrative overhead than maintaining separate databases.)

Decision: Store each tenant's resource, metric and billing data within a dedicated PostgreSQL schema while sharing a common database instance and maintaining shared public tables for common data.

Consequence:

- + Provides strong logical isolation between tenants.
- + Avoids the operational overhead of managing potentially thousands of databases.
- + Queries execute against smaller tenant-specific tables, improving index efficiency and query performance.
- + Simplifies backups, deployments, and infrastructure management compared to database-per-tenant architectures.
- + Reduces the likelihood of accidental cross-tenant queries.
- Database schema migrations must be applied to every tenant schema.
- All tenants share the same underlying database resources, requiring future horizontal scaling if database capacity is exceeded.

A.7 ADR - 007



ADR - 007

Title: Intelligence service obtaining historical context

Status: Proposed

Context: The forecasting model requires historical context to forecast values

Alternatives:

1. Open a direct connection between the intelligence service and the database.

Decision: Use the application service to query the required historical context using the existing database connection and communicate context with intelligence service via an HTTP protocol.

Consequence:

- + Reduces operational complexity, not needing to open DB connection from intelligence service.
- + Decouples the intelligence service from the database schema.
- + Reduces attack surface by not introducing additional paths to the DB that can potentially introduce high risk vulnerabilities.
- Network latency due to large payloads.
- API contract maintenance.

A.8 ADR - 008

ADR - 008

Title: Use of Spring Scheduler for Asynchronous Optimization Worker

Status: Accepted

Context: The Optimization Recommendation Engine uses an asynchronous, batch-processing architecture designed to decouple heavy analytical queries from synchronous user interactions. This Asynchronous Optimization Worker handles all heavy lifting, including computing heavy statistical summaries and saving them to SherpaDB. To track progress, the engine utilizes a watermark-based processing strategy, relying on a durable processing_watermark table to check exactly when it last successfully processed data for a specific tenant. Because state management and execution tracking are natively handled by this watermark table, the scheduling mechanism strictly needs to act as a lightweight job tracker.

Alternatives:

1. JobRunr

- **Advantages:** Provides distributed background job processing, and automatic retry mechanisms.
- **Disadvantages:** Introduces unnecessary database overhead and complexity.

2. Spring Scheduler

- **Advantages:** Lightweight, native to the Spring ecosystem, and easily configured to execute the worker once each day. It delegates tracking entirely to our database-native watermark strategy.
- **Disadvantages:** Lack of Built-In Retry and Failure Handling

Decision: We will use a standard Spring scheduler for the Optimization Worker. The worker will trigger once each day, at which point it will read the processing_watermark table to determine where it left off, pull the last 24 hours of unprocessed normalized metrics, run rule evaluations, update the recommendations, and go back to sleep.

Consequence:

- + Reduces overall architectural complexity by eliminating a heavy, third-party job tracking framework.
- + Maintains a single source of truth for processing state via the custom processing_watermark table.
- Introduces the risk of 24-hour data delays upon failure, as the scheduler lacks automatic retries for database or network errors.

A.9 ADR - 009



ADR - 009

Title: Dynamic billing ingestion bean registry via factory method vs simple switch-based dispatch logic

Status: Proposed

Context: The billing ingestion services for each provider subscribe to an interface that enforces the execute method contract that the scheduler uses to schedule ingestion.

Alternatives:

1. Use a factory pattern leveraging spring dependency injection to automatically load all services implementing this interface into a registry keyed by the provider enum. The scheduler code would then be unaware of the specific billing ingestion implementations and would be able to call the correct execute method with just the provider as context

Decision: Manually inject the billing ingestion services into the scheduler BillingIngestionClient and dispatch behavior via a switch statement on the provider enum and switch labeled rules for improved readability and explicit behavior for the limited scope of three Cloud Providers

Consequence:

- Adding a new billing ingestion service for a new provider requires manually injecting the service into the BillingIngestionClient

B | Appendix B: Service Contracts

B.1 Ingestion Service

```
1 {
2   "openapi": "3.1.0",
3   "info": {
4     "title": "OpenAPI definition",
5     "version": "v0"
6   },
7   "servers": [
8     {
9       "url": "http://localhost:8081",
10      "description": "Generated server url"
11    }
12  ],
13  "tags": [
14    {
15      "name": "Cloud Usage Ingestion",
16      "description": "Endpoints for ingesting cloud usage and
17        billing data from cloud providers"
18    },
19    {
20      "name": "Cloud Resource Discovery",
21      "description": "Endpoints for discovering which services
22        and resources are available from cloud providers"
23    }
24  ],
25  "paths": {
26    "/api/events/ingest": {
27      "post": {
28        "tags": [
29          "Cloud Usage Ingestion"
30        ],
31        "summary": "Ingest real cloud usage and billing data",
32        "description": "Retrieves cloud usage and/or billing data
33          from configured cloud providers using the supplied
34          credentials, account scopes, and time range.",
35        "operationId": "ingest",
36        "requestBody": {
```

```

33         "description": "Cloud ingestion request object",
34         "content": {
35             "application/json": {
36                 "schema": {
37                     "$ref": "#/components/schemas/
                        IngestionRequestEvent"
38                 }
39             }
40         },
41         "required": true
42     },
43     "responses": {
44         "200": {
45             "description": "Cloud data successfully ingested",
46             "content": {
47                 "application/json": {
48                     "schema": {
49                         "$ref": "#/components/schemas/IngestionResult"
50                     }
51                 }
52             }
53         }
54     }
55 },
56 "/api/events/ingest/mock": {
57     "post": {
58         "tags": [
59             "Cloud Usage Ingestion"
60         ],
61         "summary": "Ingest deterministic fixed-size mock cloud
                        data",
62         "description": "Generates and ingests deterministic mock
                        cloud usage datafor development and integration
                        testing.",
63         "operationId": "ingestMock",
64         "requestBody": {
65             "description": "Ingestion request object, credentials
                        and some other fields are not used",
66             "content": {
67                 "application/json": {
68                     "schema": {
69                         "$ref": "#/components/schemas/
                        IngestionRequestEvent"
70                     }

```

```

71         }
72     },
73     },
74     "required": true
75 },
76 "responses": {
77     "200": {
78         "description": "Mock cloud data successfully ingested",
79         "content": {
80             "application/json": {
81                 "schema": {
82                     "$ref": "#/components/schemas/IngestionResult"
83                 }
84             }
85         }
86     }
87 },
88 },
89 },
90 "/api/events/ingest/mockNoise": {
91     "post": {
92         "tags": [
93             "Cloud Usage Ingestion"
94         ],
95         "summary": "Ingest mock cloud data with noise",
96         "description": "Generates and ingests mock cloud usage data using Reinstein Uhlenbeck noise for analytics, anomaly detection, and testing.",
97         "operationId": "ingestMockWithNoise",
98         "requestBody": {
99             "description": "Ingestion request object, credentials and some other fields are not used",
100             "content": {
101                 "application/json": {
102                     "schema": {
103                         "$ref": "#/components/schemas/IngestionRequestEvent"
104                     }
105                 }
106             },
107             "required": true
108         },
109         "responses": {

```

```
110         "200": {
111             "description": "Mock cloud data successfully ingested",
112             "content": {
113                 "application/json": {
114                     "schema": {
115                         "$ref": "#/components/schemas/IngestionResult"
116                     }
117                 }
118             }
119         }
120     },
121 },
122 "/api/events/ingest/gcp/billing/bigquery": {
123     "post": {
124         "tags": [
125             "Cloud Usage Ingestion"
126         ],
127         "summary": "Trigger GCP BigQuery billing ingestion manually",
128         "description": "Runs GCP BigQuery billing ingestion for testing and development. This endpoint is only available when the dev Spring profile is active.",
129         "operationId": "ingestGcpBigqueryBilling",
130         "responses": {
131             "200": {
132                 "description": "GCP BigQuery billing ingestion ran successfully"
133             },
134             "401": {
135                 "description": "Endpoint is only available in the dev profile"
136             }
137         }
138     }
139 },
140 "/api/events/ingest/aws/billing/cur": {
141     "post": {
142         "tags": [
143             "Cloud Usage Ingestion"
144         ],
145         "summary": "Trigger AWS CUR billing ingestion with credentials stored for test user",
146     }
```

```

147     "description": "When database is seeded appropriately,
148         ingest a billing report for testing purposes",
149     "operationId": "ingestAwsBillingCur",
150     "responses": {
151         "200": {
152             "description": "AWS CUR billing ingestion ran
153                 successfully",
154             "content": {
155                 "application/json": {
156                     "schema": {
157                         "$ref": "#/components/schemas/AwsCurContext"
158                     }
159                 }
160             }
161         }
162     },
163     "/api/cloud-resources/services": {
164         "post": {
165             "tags": [
166                 "Cloud Resource Discovery"
167             ],
168             "summary": "Get all supported services for a cloud
169                 provider",
170             "description": "Retrieves a list of all services that can
171                 be monitored by CloudSherpa for a given provider",
172             "operationId": "getAllOfferedServices",
173             "requestBody": {
174                 "description": "Cloud provider name (e.g. aws, azure,
175                     gcp)",
176                 "content": {
177                     "application/json": {
178                         "schema": {
179                             "type": "string"
180                         }
181                     }
182                 },
183                 "required": true
184             },
185             "responses": {
186                 "200": {
187                     "description": "Cloud services successfully returned
188                         ",

```

```

185         "content": {
186             "application/json": {
187                 "schema": {
188                     "type": "string"
189                 }
190             }
191         }
192     }
193 }
194 },
195 {
196     "/api/cloud-resources/resources/{provider}": {
197         "post": {
198             "tags": [
199                 "Cloud Resource Discovery"
200             ],
201             "summary": "Get all resources within provided service
202                 categories for a cloud provider",
203             "description": "Retrieves all discoverable resources
204                 within a list of services of a cloud provider using
205                 the supplied credentials",
206             "operationId": "getAllResources",
207             "parameters": [
208                 {
209                     "name": "provider",
210                     "in": "path",
211                     "required": true,
212                     "schema": {
213                         "type": "string"
214                     }
215                 }
216             ],
217             "requestBody": {
218                 "content": {
219                     "application/json": {
220                         "schema": {
221                             "$ref": "#/components/schemas/
222                                 ResourceDiscoveryRequest "
223                         }
224                     }
225                 },
226                 "required": true
227             },
228             "responses": {

```

```

225         "200": {
226             "description": "Resources successfully returned",
227             "content": {
228                 "application/json": {
229                     "schema": {
230                         "$ref": "#/components/schemas/ResourceDetail"
231                     }
232                 }
233             }
234         }
235     },
236 },
237 "/api/cloud-resources/gcp/permissions": {
238     "post": {
239         "tags": [
240             "Cloud Resource Discovery"
241         ],
242         "summary": "Generate GCP permissions list",
243         "description": "Generates a least-privilege GCP
244             permissions list for the selected services",
245         "operationId": "generateGcpPermissionsPolicy",
246         "requestBody": {
247             "content": {
248                 "application/json": {
249                     "schema": {
250                         "type": "array",
251                         "items": {
252                             "type": "string"
253                         }
254                     }
255                 }
256             },
257             "required": true
258         },
259         "responses": {
260             "200": {
261                 "description": "Permissions list successfully
262                     generated",
263                 "content": {
264                     "*/*": {
265                         "schema": {
266                             "type": "array",

```

```

267         "type": "string"
268     },
269     "uniqueItems": true
270 }
271 }
272 }
273 }
274 }
275 }
276 },
277 "/api/cloud-resources/azure/permissions": {
278     "post": {
279         "tags": [
280             "Cloud Resource Discovery"
281         ],
282         "summary": "Generate Azure permissions list",
283         "description": "Generates a least-privilege Azure
284             permissions list for the selected services",
285         "operationId": "generateAzurePermissionsPolicy",
286         "requestBody": {
287             "content": {
288                 "application/json": {
289                     "schema": {
290                         "type": "array",
291                         "items": {
292                             "type": "string"
293                         }
294                     }
295                 },
296                 "required": true
297             },
298             "responses": {
299                 "200": {
300                     "description": "Permissions list successfully
301                         generated",
302                     "content": {
303                         "*/*": {
304                             "schema": {
305                                 "type": "array",
306                                 "items": {
307                                     "type": "string"
308                                 },
309                                 "uniqueItems": true

```

```

309         }
310     }
311 }
312 }
313 }
314 }
315 },
316 "/api/cloud-resources/aws/permissions": {
317     "post": {
318         "tags": [
319             "Cloud Resource Discovery"
320         ],
321         "summary": "Generate AWS IAM policy",
322         "description": "Generates a least-privilege AWS IAM
323             policy for the selected services",
324         "operationId": "generateAwsPermissionsPolicy",
325         "requestBody": {
326             "content": {
327                 "application/json": {
328                     "schema": {
329                         "type": "array",
330                         "items": {
331                             "type": "string"
332                         }
333                     }
334                 },
335                 "required": true
336             },
337             "responses": {
338                 "200": {
339                     "description": "IAM policy successfully generated",
340                     "content": {
341                         "*/*": {
342                             "schema": {
343                                 "type": "string"
344                             }
345                         }
346                     }
347                 }
348             }
349         }
350     }
351 },

```

```
352 "components": {
353   "schemas": {
354     "AccountScope": {
355       "type": "object",
356       "properties": {
357         "provider": {
358           "type": "string"
359         },
360         "accountId": {
361           "type": "string"
362         },
363         "subscriptionId": {
364           "type": "string"
365         },
366         "projectId": {
367           "type": "string"
368         },
369         "billingAccountId": {
370           "type": "string"
371         },
372         "serviceScopes": {
373           "type": "array",
374           "items": {
375             "$ref": "#/components/schemas/ServiceScope"
376           }
377         }
378       }
379     },
380     "CloudCredentials": {
381       "type": "object",
382       "properties": {
383         "accessKeyId": {
384           "type": "string"
385         },
386         "secretAccessKey": {
387           "type": "string"
388         },
389         "subscriptionId": {
390           "type": "string"
391         },
392         "tenantId": {
393           "type": "string"
394         },
395         "clientId": {
```

```
396         "type": "string"
397     },
398     "clientSecret": {
399         "type": "string"
400     },
401     "projectId": {
402         "type": "string"
403     },
404     "serviceAccountJson": {
405         "type": "string"
406     }
407 }
408 },
409 "IngestionRequestEvent": {
410     "type": "object",
411     "properties": {
412         "scopes": {
413             "type": "array",
414             "items": {
415                 "$ref": "#/components/schemas/AccountScope"
416             }
417         },
418         "credentials": {
419             "$ref": "#/components/schemas/CloudCredentials"
420         },
421         "from": {
422             "type": "string",
423             "format": "date-time"
424         },
425         "to": {
426             "type": "string",
427             "format": "date-time"
428         },
429         "period": {
430             "type": "integer",
431             "format": "int32"
432         },
433         "includeBilling": {
434             "type": "boolean"
435         },
436         "includeUsage": {
437             "type": "boolean"
438         },
439         "userId": {
```

```
440         "type": "string",
441         "format": "uuid"
442     }
443 }
444 },
445 "Instance": {
446     "type": "object",
447     "properties": {
448         "identifier": {
449             "type": "string"
450         },
451         "region": {
452             "type": "string"
453         }
454     }
455 },
456 "InstanceScope": {
457     "type": "object",
458     "properties": {
459         "identifierName": {
460             "type": "string"
461         },
462         "instances": {
463             "type": "array",
464             "items": {
465                 "$ref": "#/components/schemas/Instance"
466             }
467         }
468     }
469 },
470 "Metric": {
471     "type": "object",
472     "properties": {
473         "name": {
474             "type": "string"
475         },
476         "unit": {
477             "type": "string"
478         }
479     }
480 },
481 "ServiceScope": {
482     "type": "object",
483     "properties": {
```

```
484     "name": {
485         "type": "string"
486     },
487     "instances": {
488         "type": "array",
489         "items": {
490             "$ref": "#/components/schemas/InstanceScope"
491         }
492     },
493     "metrics": {
494         "type": "array",
495         "items": {
496             "$ref": "#/components/schemas/Metric"
497         }
498     }
499 },
500 "BillingRecordModel": {
501     "type": "object",
502     "properties": {
503         "recordId": {
504             "type": "string",
505             "format": "uuid"
506         },
507         "provider": {
508             "type": "string"
509         },
510         "accountId": {
511             "type": "string"
512         },
513         "subscriptionId": {
514             "type": "string"
515         },
516         "projectId": {
517             "type": "string"
518         },
519         "billingAccountId": {
520             "type": "string"
521         },
522         "serviceName": {
523             "type": "string"
524         },
525         "resourceId": {
526             "type": "string"
527         }
```

```
528     },
529     "resourceType": {
530       "type": "string"
531     },
532     "region": {
533       "type": "string"
534     },
535     "cost": {
536       "type": "number",
537       "format": "double"
538     },
539     "usageQuantity": {
540       "type": "number",
541       "format": "double"
542     },
543     "unit": {
544       "type": "string"
545     },
546     "currency": {
547       "type": "string"
548     },
549     "pricingModel": {
550       "type": "string"
551     },
552     "usageStartTime": {
553       "type": "string",
554       "format": "date-time"
555     },
556     "usageEndTime": {
557       "type": "string",
558       "format": "date-time"
559     },
560     "billingPeriodStart": {
561       "type": "string",
562       "format": "date-time"
563     },
564     "billingPeriodEnd": {
565       "type": "string",
566       "format": "date-time"
567     },
568     "tags": {
569       "type": "object",
570       "additionalProperties": {
571         "type": "string"
```

```
572     }
573   },
574   "ingestionTimestamp": {
575     "type": "string",
576     "format": "date-time"
577   },
578   "ingestionId": {
579     "type": "string"
580   },
581   "source": {
582     "type": "string"
583   }
584 }
585 },
586 "IngestionResult": {
587   "type": "object",
588   "properties": {
589     "usage": {
590       "type": "array",
591       "items": {
592         "$ref": "#/components/schemas/UsageRecordModel"
593       }
594     },
595     "billing": {
596       "type": "array",
597       "items": {
598         "$ref": "#/components/schemas/BillingRecordModel"
599       }
600     }
601   }
602 },
603 "UsageRecordModel": {
604   "type": "object",
605   "properties": {
606     "recordId": {
607       "type": "string",
608       "format": "uuid"
609     },
610     "provider": {
611       "type": "string"
612     },
613     "accountId": {
614       "type": "string"
615     },

```

```
616     "subscriptionId": {
617         "type": "string"
618     },
619     "projectId": {
620         "type": "string"
621     },
622     "resourceId": {
623         "type": "string"
624     },
625     "resourceType": {
626         "type": "string"
627     },
628     "serviceName": {
629         "type": "string"
630     },
631     "region": {
632         "type": "string"
633     },
634     "metricName": {
635         "type": "string"
636     },
637     "value": {
638         "type": "number",
639         "format": "double"
640     },
641     "unit": {
642         "type": "string"
643     },
644     "timestamp": {
645         "type": "string",
646         "format": "date-time"
647     },
648     "periodStart": {
649         "type": "string",
650         "format": "date-time"
651     },
652     "periodEnd": {
653         "type": "string",
654         "format": "date-time"
655     },
656     "dimensions": {
657         "type": "object",
658         "additionalProperties": {
659             "type": "string"
```

```
660     }
661   },
662   "tags": {
663     "type": "object",
664     "additionalProperties": {
665       "type": "string"
666     }
667   },
668   "ingestionTimestamp": {
669     "type": "string",
670     "format": "date-time"
671   },
672   "ingestionId": {
673     "type": "string"
674   },
675   "source": {
676     "type": "string"
677   }
678 },
679 "AwsCurContext": {
680   "type": "object",
681   "properties": {
682     "bucketName": {
683       "type": "string"
684     },
685     "bucketRegion": {
686       "$ref": "#/components/schemas/Region"
687     },
688     "exportPrefix": {
689       "type": "string"
690     },
691     "exportName": {
692       "type": "string"
693     },
694     "accountId": {
695       "type": "string",
696       "format": "uuid"
697     },
698     "processedExports": {
699       "type": "array",
700       "items": {
701         "type": "string"
702       }
703     }
```

```
704     },
705     "configId": {
706       "type": "string"
707     },
708     "userId": {
709       "type": "string"
710     },
711     "dataFiles": {
712       "type": "array",
713       "items": {
714         "type": "string"
715       }
716     },
717     "awsCurTmpDir": {
718       "type": "object",
719       "properties": {
720         "absolute": {
721           "type": "boolean"
722         },
723         "parent": {
724           "type": "object",
725           "properties": {
726             "absolute": {
727               "type": "boolean"
728             },
729             "root": {
730               "type": "object",
731               "properties": {
732                 "absolute": {
733                   "type": "boolean"
734                 },
735                 "fileName": {
736                   "type": "object",
737                   "properties": {
738                     "absolute": {
739                       "type": "boolean"
740                     },
741                     "fileSystem": {
742                       "type": "object",
743                       "properties": {
744                         "open": {
745                           "type": "boolean"
746                         },
747                         "readOnly": {
```

```
748         "type": "boolean"
749     },
750     "separator": {
751         "type": "string"
752     },
753     "rootDirectories": {
754
755     },
756     "fileStores": {
757
758     },
759     "userPrincipalLookupService": {
760
761     }
762 }
763 },
764 "nameCount": {
765     "type": "integer",
766     "format": "int32"
767 }
768 }
769 },
770 "fileSystem": {
771     "type": "object",
772     "properties": {
773         "open": {
774             "type": "boolean"
775         },
776         "readOnly": {
777             "type": "boolean"
778         },
779         "separator": {
780             "type": "string"
781         },
782         "rootDirectories": {
783
784         },
785         "fileStores": {
786
787         },
788         "userPrincipalLookupService": {
789
790         }
791     }
```

```
792         },
793         "nameCount": {
794             "type": "integer",
795             "format": "int32"
796         }
797     },
798     "fileName": {
799         "type": "object",
800         "properties": {
801             "absolute": {
802                 "type": "boolean"
803             },
804             "fileSystem": {
805                 "type": "object",
806                 "properties": {
807                     "open": {
808                         "type": "boolean"
809                     },
810                     "readOnly": {
811                         "type": "boolean"
812                     },
813                     "separator": {
814                         "type": "string"
815                     },
816                     "rootDirectories": {
817
818                     },
819                     "fileStores": {
820
821                     },
822                     "userPrincipalLookupService": {
823
824                     }
825                 }
826             },
827             "nameCount": {
828                 "type": "integer",
829                 "format": "int32"
830             }
831         }
832     },
833     "fileSystem": {
834         "type": "object",
```

```
836         "properties": {
837             "open": {
838                 "type": "boolean"
839             },
840             "readOnly": {
841                 "type": "boolean"
842             },
843             "separator": {
844                 "type": "string"
845             },
846             "rootDirectories": {
847
848             },
849             "fileStores": {
850
851             },
852             "userPrincipalLookupService": {
853
854             }
855         },
856     },
857     "nameCount": {
858         "type": "integer",
859         "format": "int32"
860     }
861 }
862 },
863 "root": {
864     "type": "object",
865     "properties": {
866         "absolute": {
867             "type": "boolean"
868         },
869         "fileName": {
870             "type": "object",
871             "properties": {
872                 "absolute": {
873                     "type": "boolean"
874                 },
875                 "fileSystem": {
876                     "type": "object",
877                     "properties": {
878                         "open": {
879                             "type": "boolean"
```

```
880         },
881         "readOnly": {
882             "type": "boolean"
883         },
884         "separator": {
885             "type": "string"
886         },
887         "rootDirectories": {
888
889         },
890         "fileStores": {
891
892         },
893         "userPrincipalLookupService": {
894
895         }
896     },
897     },
898     "nameCount": {
899         "type": "integer",
900         "format": "int32"
901     }
902 },
903 },
904 "fileSystem": {
905     "type": "object",
906     "properties": {
907         "open": {
908             "type": "boolean"
909         },
910         "readOnly": {
911             "type": "boolean"
912         },
913         "separator": {
914             "type": "string"
915         },
916         "rootDirectories": {
917
918         },
919         "fileStores": {
920
921         },
922         "userPrincipalLookupService": {
```

```
924         }
925     },
926     },
927     "nameCount": {
928         "type": "integer",
929         "format": "int32"
930     }
931 },
932 },
933 "fileName": {
934     "type": "object",
935     "properties": {
936         "absolute": {
937             "type": "boolean"
938         },
939         "fileSystem": {
940             "type": "object",
941             "properties": {
942                 "open": {
943                     "type": "boolean"
944                 },
945                 "readOnly": {
946                     "type": "boolean"
947                 },
948                 "separator": {
949                     "type": "string"
950                 },
951                 "rootDirectories": {
952
953                 },
954                 "fileStores": {
955
956                 },
957                 "userPrincipalLookupService": {
958
959                 }
960             }
961         },
962         "nameCount": {
963             "type": "integer",
964             "format": "int32"
965         }
966     }
967 },
```

```
968         "fileSystem": {
969             "type": "object",
970             "properties": {
971                 "open": {
972                     "type": "boolean"
973                 },
974                 "readOnly": {
975                     "type": "boolean"
976                 },
977                 "separator": {
978                     "type": "string"
979                 },
980                 "rootDirectories": {
981
982                 },
983                 "fileStores": {
984
985                 },
986                 "userPrincipalLookupService": {
987
988                 }
989             }
990         },
991         "nameCount": {
992             "type": "integer",
993             "format": "int32"
994         }
995     },
996     "userUuid": {
997         "type": "string",
998         "format": "uuid"
999     }
1000 }
1001 }
1002 },
1003 "Region": {
1004     "type": "object",
1005     "properties": {
1006         "globalRegion": {
1007             "type": "boolean"
1008         }
1009     }
1010 },
1011 "ResourceDiscoveryRequest": {
```

```
1012     "type": "object",
1013     "properties": {
1014         "services": {
1015             "type": "array",
1016             "items": {
1017                 "type": "string"
1018             }
1019         },
1020         "credentials": {
1021             "$ref": "#/components/schemas/CloudCredentials"
1022         }
1023     }
1024 },
1025 "ResourceDetail": {
1026     "type": "object",
1027     "properties": {
1028         "resourceId": {
1029             "type": "string"
1030         },
1031         "name": {
1032             "type": "string"
1033         },
1034         "resourceType": {
1035             "type": "string"
1036         },
1037         "serviceCategory": {
1038             "type": "string"
1039         },
1040         "region": {
1041             "type": "string"
1042         },
1043         "tags": {
1044             "type": "object",
1045             "additionalProperties": {
1046                 "type": "string"
1047             }
1048         }
1049     }
1050 }
1051 }
1052 }
1053 }
```

B.2 Service

```
1 {
2   "openapi": "3.1.0",
3   "info": {
4     "title": "OpenAPI definition",
5     "version": "v0"
6   },
7   "servers": [
8     {
9       "url": "http://localhost:8083",
10      "description": "Generated server url"
11    }
12  ],
13  "tags": [
14    {
15      "name": "Authentication",
16      "description": "Endpoints for user registration and login"
17    },
18    {
19      "name": "CloudSherpa Analytics",
20      "description": "CloudSherpa Analytics endpoints"
21    },
22    {
23      "name": "Optimization Recommendations",
24      "description": "Retrieve and manage cloud optimization
25        recommendations"
26    },
27    {
28      "name": "Intelligence",
29      "description": "CloudSherpa Forecasting Intelligence
30        Operations"
31    },
32    {
33      "name": "CloudSherpa Dashboard",
34      "description": "CloudSherpa Dashboard endpoints"
35    },
36    {
37      "name": "Billing",
38      "description": "Billing KPI endpoints"
39    },
40    {
41      "name": "Cloud Resource Discovery",
```

```

40     "description": "Endpoints for discovering cloud services,
41         resources, and generating permissions policies"
42     },
43     "paths": {
44         "/dashboards/{dashboardId}/layout": {
45             "put": {
46                 "tags": [
47                     "CloudSherpa Dashboard"
48                 ],
49                 "summary": "Batch update the sizes and positions of all
50                     widgets on a dashboard",
51                 "operationId": "updateDashboardLayout",
52                 "parameters": [
53                     {
54                         "name": "dashboardId",
55                         "in": "path",
56                         "required": true,
57                         "schema": {
58                             "type": "string",
59                             "format": "uuid"
60                         }
61                     }
62                 ],
63                 "requestBody": {
64                     "content": {
65                         "application/json": {
66                             "schema": {
67                                 "type": "array",
68                                 "items": {
69                                     "$ref": "#/components/schemas/
70                                         WidgetLayoutUpdatedDTO"
71                                 }
72                             }
73                         },
74                         "required": true
75                     },
76                     "responses": {
77                         "200": {
78                             "description": "OK"
79                         }
80                     }
81                 }
82             }
83         }
84     }
85 }

```

```
81     },
82     "/preferences/theme": {
83         "get": {
84             "tags": [
85                 "preferences-controller"
86             ],
87             "operationId": "getTheme",
88             "responses": {
89                 "200": {
90                     "description": "OK",
91                     "content": {
92                         "*/*": {
93                             "schema": {
94                                 "type": "object",
95                                 "additionalProperties": {
96                                     "type": "string"
97                                 }
98                             }
99                         }
100                     }
101                 }
102             }
103         },
104         "post": {
105             "tags": [
106                 "preferences-controller"
107             ],
108             "operationId": "updateTheme",
109             "requestBody": {
110                 "content": {
111                     "application/json": {
112                         "schema": {
113                             "$ref": "#/components/schemas/ThemeUpdateRequest"
114                         }
115                     }
116                 },
117                 "required": true
118             },
119             "responses": {
120                 "200": {
121                     "description": "OK"
122                 }
123             }
124         }
125     }
```

```
125 },
126 "/optimization/recommendations/{recommendationId}/re-enable":
127 {
128   "post": {
129     "tags": [
130       "Optimization Recommendations"
131     ],
132     "operationId": "reEnableRecommendation",
133     "parameters": [
134       {
135         "name": "recommendationId",
136         "in": "path",
137         "required": true,
138         "schema": {
139           "type": "string",
140           "format": "uuid"
141         }
142       }
143     ],
144     "responses": {
145       "200": {
146         "description": "OK",
147         "content": {
148           "*/*": {
149             "schema": {
150               "type": "object",
151               "additionalProperties": {}
152             }
153           }
154         }
155       }
156     }
157   },
158   "/optimization/recommendations/{recommendationId}/dismiss": {
159     "post": {
160       "tags": [
161         "Optimization Recommendations"
162       ],
163       "summary": "Dismiss a recommendation",
164       "description": "Marks an active recommendation as dismissed.",
165       "operationId": "dismissRecommendation",
166       "parameters": [
```

```

167     {
168         "name": "recommendationId",
169         "in": "path",
170         "description": "Recommendation UUID",
171         "required": true,
172         "schema": {
173             "type": "string",
174             "format": "uuid"
175         }
176     }
177 ],
178 "responses": {
179     "200": {
180         "description": "OK",
181         "content": {
182             "*/*": {
183                 "schema": {
184                     "type": "object",
185                     "additionalProperties": {}
186                 }
187             }
188         }
189     }
190 }
191 },
192 "/optimization/recommendations/{recommendationId}/apply": {
193     "post": {
194         "tags": [
195             "Optimization Recommendations"
196         ],
197         "summary": "Apply a recommendation",
198         "description": "Marks an active recommendation as applied",
199         ".",
200         "operationId": "applyRecommendation",
201         "parameters": [
202             {
203                 "name": "recommendationId",
204                 "in": "path",
205                 "description": "Recommendation UUID",
206                 "required": true,
207                 "schema": {
208                     "type": "string",
209                     "format": "uuid"

```

```

210     }
211   }
212 ],
213   "responses": {
214     "200": {
215       "description": "OK",
216       "content": {
217         "*/*": {
218           "schema": {
219             "type": "object",
220             "additionalProperties": {}
221           }
222         }
223       }
224     }
225   }
226 },
227 "/optimization/recommendations/{recommendationId}/acknowledge
228   ": {
229     "post": {
230       "tags": [
231         "Optimization Recommendations"
232       ],
233       "summary": "Acknowledge a recommendation",
234       "description": "Marks an active recommendation as
235         acknowledged.",
236       "operationId": "acknowledgeRecommendation",
237       "parameters": [
238         {
239           "name": "recommendationId",
240           "in": "path",
241           "description": "Recommendation UUID",
242           "required": true,
243           "schema": {
244             "type": "string",
245             "format": "uuid"
246           }
247         }
248       ],
249       "responses": {
250         "200": {
251           "description": "OK",
252           "content": {

```

```

252         "*/*": {
253             "schema": {
254                 "type": "object",
255                 "additionalProperties": {}
256             }
257         }
258     }
259 }
260 }
261 }
262 },
263 "/intelligence/forecasting/resource": {
264     "post": {
265         "tags": [
266             "Intelligence"
267         ],
268         "summary": "Forecast resource usage",
269         "description": "Generates predicted values and prediction
                          intervals for a resource metric up to the requested
                          forecast horizon",
270         "operationId": "resourceUsageForecast",
271         "requestBody": {
272             "description": "The resource, metric, and timestamp
                          through which usage should be forecast",
273             "content": {
274                 "application/json": {
275                     "schema": {
276                         "$ref": "#/components/schemas/
                          ResourceUsageForecastRequestDto"
277                     }
278                 }
279             },
280             "required": true
281         },
282         "responses": {
283             "200": {
284                 "description": "Resource usage forecast generated
                          successfully",
285                 "content": {
286                     "application/json": {
287                         "schema": {
288                             "$ref": "#/components/schemas/
                          ResourceUsageForecastResponseDto"
289                         }

```

```

290         }
291     },
292     "400": {
293         "description": "The forecast request is malformed or
294             contains invalid values"
295     },
296     "422": {
297         "description": "Insufficient historical data
298             available to make forecasting prediction"
299     },
300     "404": {
301         "description": "Resource does not exist or MetricType
302             is not applicable to Resource"
303     }
304 },
305 "/intelligence/forecasting/billing": {
306     "post": {
307         "tags": [
308             "Intelligence"
309         ],
310         "summary": "Forecast Billing for all non-credit charges",
311         "description": "Generates cumalative cost predection
312             value and analytics for all non-credit charges",
313         "operationId": "billingForecast",
314         "requestBody": {
315             "content": {
316                 "application/json": {
317                     "schema": {
318                         "$ref": "#/components/schemas/
319                             BillingForecastRequest"
320                     }
321                 },
322                 "required": true
323             },
324             "responses": {
325                 "200": {
326                     "description": "Billing forecast generated
327                         succesfully.",
328                     "content": {
329                         "application/json": {

```

```

328         "schema": {
329             "$ref": "#/components/schemas/
330                 BillingForecastResponseDto"
331         }
332     },
333     "422": {
334         "description": "Insufficient historical data
335             available to make any forecasting prediction"
336     },
337     "404": {
338         "description": "No non-credit charges found"
339     }
340 }
341 },
342 "/intelligence/forecasting/billing-charges": {
343     "post": {
344         "tags": [
345             "Intelligence"
346         ],
347         "summary": "Forecast Billing with Charges",
348         "description": "Generates cumalative cost predection
349             value and analytics for list of charges of which the
350             List can contain a single or multiple charges",
351         "operationId": "billingForecastWithCharges",
352         "requestBody": {
353             "content": {
354                 "application/json": {
355                     "schema": {
356                         "$ref": "#/components/schemas/
357                             BillingForecastIndividualChargesRequestDto"
358                     }
359                 }
360             },
361             "required": true
362         },
363         "responses": {
364             "200": {
365                 "description": "Billing forecast generated
366                     succesfully.",
367                 "content": {
368                     "application/json": {

```

```

366         "schema": {
367             "$ref": "#/components/schemas/
                 BillingForecastResponseDto"
368         }
369     }
370 }
371 },
372 "400": {
373     "description": "The billing forecast request is
                 malformed or contains invalid values"
374 },
375 "422": {
376     "description": "Insufficient historical data
                 available to make any forecasting prediction"
377 },
378 "404": {
379     "description": "None of the charges found"
380 }
381 }
382 }
383 },
384 "/gcp/connections": {
385     "get": {
386         "tags": [
387             "connection-query-controller"
388         ],
389         "operationId": "getAccountConnections",
390         "responses": {
391             "200": {
392                 "description": "OK",
393                 "content": {
394                     "*/*": {
395                         "schema": {
396                             "type": "array",
397                             "items": {
398                                 "$ref": "#/components/schemas/CloudAccount"
399                             }
400                         }
401                     }
402                 }
403             }
404         }
405     },
406     "post": {

```

```

407     "tags": [
408         "gcp-connection-persistence-controller"
409     ],
410     "operationId": "persistConnection",
411     "requestBody": {
412         "content": {
413             "application/json": {
414                 "schema": {
415                     "$ref": "#/components/schemas/
                        PersistGcpConnectionRequest"
416                 }
417             }
418         },
419         "required": true
420     },
421     "responses": {
422         "200": {
423             "description": "OK"
424         }
425     }
426 },
427 "/dashboards": {
428     "get": {
429         "tags": [
430             "CloudSherpa Dashboard"
431         ],
432         "summary": "Get all user dashboards with their widgets",
433         "operationId": "getUserDashboards",
434         "responses": {
435             "200": {
436                 "description": "OK",
437                 "content": {
438                     "*/*": {
439                         "schema": {
440                             "type": "array",
441                             "items": {
442                                 "$ref": "#/components/schemas/DashboardDTO"
443                             }
444                         }
445                     }
446                 }
447             }
448         }
449     }

```

```

450     },
451     "post": {
452         "tags": [
453             "CloudSherpa Dashboard"
454         ],
455         "summary": "Create a new blank dashboard",
456         "operationId": "createDashboard",
457         "requestBody": {
458             "content": {
459                 "application/json": {
460                     "schema": {
461                         "$ref": "#/components/schemas/DashboardCreateDTO"
462                     }
463                 }
464             },
465             "required": true
466         },
467         "responses": {
468             "201": {
469                 "description": "Created",
470                 "content": {
471                     "*/*": {
472                         "schema": {
473                             "$ref": "#/components/schemas/DashboardDTO"
474                         }
475                     }
476                 }
477             }
478         }
479     },
480     "/dashboards/{dashboardId}/widgets": {
481         "post": {
482             "tags": [
483                 "CloudSherpa Dashboard"
484             ],
485             "summary": "Add a new widget to a dashboard",
486             "operationId": "createChartWidget",
487             "parameters": [
488                 {
489                     "name": "dashboardId",
490                     "in": "path",
491                     "required": true,
492                     "schema": {

```

```
494         "type": "string",
495         "format": "uuid"
496     }
497 }
498 ],
499 "requestBody": {
500     "content": {
501         "application/json": {
502             "schema": {
503                 "oneOf": [
504                     {
505                         "$ref": "#/components/schemas/ChartWidgetDTO"
506                     },
507                     {
508                         "$ref": "#/components/schemas/KpiWidgetDTO"
509                     }
510                 ]
511             }
512         }
513     },
514     "required": true
515 },
516 "responses": {
517     "201": {
518         "description": "Created",
519         "content": {
520             "*/*": {
521                 "schema": {
522                     "$ref": "#/components/schemas/WidgetDTO"
523                 }
524             }
525         }
526     }
527 }
528 },
529 "/billing/kpis/preview": {
530     "post": {
531         "tags": [
532             "Billing"
533         ],
534         "summary": "Preview billing KPI card",
535         "description": "Returns a preview value for a KPI card
536             using selected resources, date range, and aggregation
```

```

537         ",
538         "operationId": "previewKpi",
539         "requestBody": {
540             "content": {
541                 "application/json": {
542                     "schema": {
543                         "$ref": "#/components/schemas/BillingKpiRequest"
544                     }
545                 },
546                 "required": true
547             },
548             "responses": {
549                 "200": {
550                     "description": "KPI preview generated",
551                     "content": {
552                         "*/*": {
553                             "schema": {
554                                 "$ref": "#/components/schemas/
BillingKpiResponse"
555                             }
556                         }
557                     },
558                     "400": {
559                         "description": "Invalid request",
560                         "content": {
561                             "*/*": {
562                                 "schema": {
563                                     "$ref": "#/components/schemas/
BillingKpiResponse"
564                                 }
565                             }
566                         }
567                     }
568                 }
569             }
570         },
571         "/azure/connections": {
572             "get": {
573                 "tags": [
574                     "connection-query-controller"
575                 ],
576                 "operationId": "getAccountConnections_1",
577

```

```
578     "responses": {
579       "200": {
580         "description": "OK",
581         "content": {
582           "*/*": {
583             "schema": {
584               "type": "array",
585               "items": {
586                 "$ref": "#/components/schemas/CloudAccount"
587               }
588             }
589           }
590         }
591       }
592     },
593   },
594   "post": {
595     "tags": [
596       "azure-connection-persistence-controller"
597     ],
598     "operationId": "persistConnection_1",
599     "requestBody": {
600       "content": {
601         "application/json": {
602           "schema": {
603             "$ref": "#/components/schemas/
604               PersistAzureConnectionRequest"
605           }
606         }
607       },
608       "required": true
609     },
610     "responses": {
611       "200": {
612         "description": "OK"
613       }
614     }
615   },
616   "/aws/connections": {
617     "get": {
618       "tags": [
619         "connection-query-controller"
620       ],
```

```
621     "operationId": "getAccountConnections_2",
622     "responses": {
623       "200": {
624         "description": "OK",
625         "content": {
626           "*/*": {
627             "schema": {
628               "type": "array",
629               "items": {
630                 "$ref": "#/components/schemas/CloudAccount"
631               }
632             }
633           }
634         }
635       }
636     },
637   },
638   "post": {
639     "tags": [
640       "aws-connection-persistence-controller"
641     ],
642     "operationId": "persistConnection_2",
643     "requestBody": {
644       "content": {
645         "application/json": {
646           "schema": {
647             "$ref": "#/components/schemas/
648               PersistAwsConnectionRequest"
649           }
650         }
651       },
652       "required": true
653     },
654     "responses": {
655       "200": {
656         "description": "OK"
657       }
658     }
659   },
660   "/auth/register": {
661     "post": {
662       "tags": [
663         "Authentication"
```

```
664     ],
665     "summary": "Register a new user",
666     "operationId": "register",
667     "requestBody": {
668       "content": {
669         "application/json": {
670           "schema": {
671             "$ref": "#/components/schemas/RegisterRequest"
672           }
673         }
674       },
675       "required": true
676     },
677     "responses": {
678       "201": {
679         "description": "User successfully registered"
680       },
681       "400": {
682         "description": "Invalid input"
683       },
684       "409": {
685         "description": "Email already in use"
686       }
687     }
688   },
689   "/auth/logout": {
690     "post": {
691       "tags": [
692         "Authentication"
693       ],
694       "summary": "Logout user",
695       "operationId": "logout",
696       "responses": {
697         "200": {
698           "description": "Successfully logged out"
699         }
700       }
701     }
702   },
703   "/auth/login": {
704     "post": {
705       "tags": [
706         "Authentication"
```

```
708 ],
709 "summary": "Login user",
710 "operationId": "login",
711 "requestBody": {
712   "content": {
713     "application/json": {
714       "schema": {
715         "$ref": "#/components/schemas/LoginRequest"
716       }
717     }
718   },
719   "required": true
720 },
721 "responses": {
722   "200": {
723     "description": "Successfully authenticated",
724     "content": {
725       "*/*": {
726         "schema": {
727           "$ref": "#/components/schemas/AuthUserResponse"
728         }
729       }
730     }
731   },
732   "400": {
733     "description": "Email or password is missing",
734     "content": {
735       "*/*": {
736         "schema": {
737           "$ref": "#/components/schemas/AuthUserResponse"
738         }
739       }
740     }
741   },
742   "401": {
743     "description": "Invalid credentials",
744     "content": {
745       "*/*": {
746         "schema": {
747           "$ref": "#/components/schemas/AuthUserResponse"
748         }
749       }
750     }
751   }
752 }
```

```

752     }
753   },
754   "/api/cloud-resources/services": {
755     "post": {
756       "tags": [
757         "Cloud Resource Discovery"
758       ],
759       "summary": "Get available cloud services",
760       "description": "Retrieves a list of all supported
761         services for a given cloud provider",
762       "operationId": "getServices",
763       "requestBody": {
764         "description": "Cloud provider name (e.g. aws, azure,
765           gcp)",
766         "content": {
767           "application/json": {
768             "schema": {
769               "type": "string"
770             }
771           },
772           "required": true
773         },
774         "responses": {
775           "200": {
776             "description": "Services successfully returned",
777             "content": {
778               "application/json": {
779                 "schema": {
780                   "type": "string"
781                 }
782             }
783           }
784         }
785       }
786     },
787     "/api/cloud-resources/resources/{provider}": {
788       "post": {
789         "tags": [
790           "Cloud Resource Discovery"
791         ],
792         "summary": "Get cloud resources",

```

```
794     "description": "Retrieves all discoverable resources for
      the specified cloud provider using the supplied
      credentials",
795     "operationId": "getResources",
796     "parameters": [
797     {
798       "name": "provider",
799       "in": "path",
800       "required": true,
801       "schema": {
802         "type": "string"
803       }
804     }
805   ],
806   "requestBody": {
807     "content": {
808       "application/json": {
809         "schema": {
810           "$ref": "#/components/schemas/
            ResourceDiscoveryDto"
811         }
812       }
813     },
814     "required": true
815   },
816   "responses": {
817     "200": {
818       "description": "Resources successfully returned",
819       "content": {
820         "application/json": {
821           "schema": {
822             "$ref": "#/components/schemas/ResourceDetailDto"
823           }
824         }
825       }
826     }
827   }
828 },
829 "/api/cloud-resources/gcp/permissions": {
830   "post": {
831     "tags": [
832       "Cloud Resource Discovery"
```

```

834     ],
835     "summary": "Generate GCP permissions list",
836     "description": "Generates a least-privilege GCP
837         permissions list for the selected GCP services",
838     "operationId": "generateGcpPermissions",
839     "requestBody": {
840         "content": {
841             "application/json": {
842                 "schema": {
843                     "type": "array",
844                     "items": {
845                         "type": "string"
846                     }
847                 }
848             },
849             "required": true
850         },
851         "responses": {
852             "200": {
853                 "description": "Permissions list successfully
854                     generated",
855                 "content": {
856                     "application/json": {
857                         "schema": {
858                             "type": "string"
859                         }
860                     }
861                 }
862             }
863         }
864     },
865     "/api/cloud-resources/azure/permissions": {
866         "post": {
867             "tags": [
868                 "Cloud Resource Discovery"
869             ],
870             "summary": "Generate Azure permissions list",
871             "description": "Generates a least-privilege Azure
872                 permissions list for the selected Azure services",
873             "operationId": "generateAzurePermissions",
874             "requestBody": {
875                 "content": {

```

```
875         "application/json": {
876             "schema": {
877                 "type": "array",
878                 "items": {
879                     "type": "string"
880                 }
881             }
882         },
883     },
884     "required": true
885 },
886 "responses": {
887     "200": {
888         "description": "Permissions list successfully
889             generated",
889         "content": {
890             "application/json": {
891                 "schema": {
892                     "type": "string"
893                 }
894             }
895         }
896     }
897 },
898 },
899 },
900 "/api/cloud-resources/aws/permissions": {
901     "post": {
902         "tags": [
903             "Cloud Resource Discovery"
904         ],
905         "summary": "Generate AWS IAM policy",
906         "description": "Generates a least-privilege AWS IAM
907             policy for the selected AWS services",
908         "operationId": "generateAwsPermissions",
909         "requestBody": {
910             "content": {
911                 "application/json": {
912                     "schema": {
913                         "type": "array",
914                         "items": {
915                             "type": "string"
916                         }
917                     }
918                 }
919             }
920         }
921     }
922 }
```

```

917     }
918   },
919   "required": true
920 },
921 "responses": {
922   "200": {
923     "description": "IAM policy successfully generated",
924     "content": {
925       "application/json": {
926         "schema": {
927           "type": "string"
928         }
929       }
930     }
931   }
932 }
933 },
934 },
935 "/analytics/historical-resource-metric": {
936   "post": {
937     "tags": [
938       "CloudSherpa Analytics"
939     ],
940     "summary": "Get historical metric data for a resource",
941     "operationId": "postMethodName",
942     "requestBody": {
943       "description": "Resource metric query parameters",
944       "content": {
945         "application/json": {
946           "schema": {
947             "$ref": "#/components/schemas/ResourceMetricHistoricalRequestDto"
948           }
949         }
950       },
951       "required": true
952     },
953     "responses": {
954       "200": {
955         "description": "Return metric values and timestamps for the resource",
956         "content": {
957           "*/*": {
958             "schema": {

```

```

959         "$ref": "#/components/schemas/
           ResourceMetricHistoricalResponseDto"
960     }
961 }
962 }
963 },
964 "400": {
965     "description": "Invalid request"
966 },
967 "404": {
968     "description": "No resource metrics found"
969 }
970 }
971 }
972 },
973 "/analytics/downsampled-historical-series": {
974     "post": {
975         "tags": [
976             "CloudSherpa Analytics"
977         ],
978         "summary": "Get downsampled historical metric series for
           a specific resource and metric",
979         "operationId": "postMethodName_1",
980         "requestBody": {
981             "description": "POST parameters for downsampled
           historical metric series",
982             "content": {
983                 "application/json": {
984                     "schema": {
985                         "$ref": "#/components/schemas/
           DownsampledSeriesRequestDto"
986                     }
987                 }
988             },
989             "required": true
990         },
991         "responses": {
992             "200": {
993                 "description": "Succesfully returns downsampled
           historical metric series",
994                 "content": {
995                     "*/*": {
996                         "schema": {
997                             "type": "array",

```

```
998         "items": {
999             "$ref": "#/components/schemas/
1000                 NormalizedMetrics"
1001         }
1002     }
1003 }
1004 }
1005 }
1006 }
1007 },
1008 "/dashboards/{dashboardId}/window": {
1009     "patch": {
1010         "tags": [
1011             "CloudSherpa Dashboard"
1012         ],
1013         "operationId": "updateDashboardWindow",
1014         "parameters": [
1015             {
1016                 "name": "dashboardId",
1017                 "in": "path",
1018                 "required": true,
1019                 "schema": {
1020                     "type": "string",
1021                     "format": "uuid"
1022                 }
1023             }
1024         ],
1025         "requestBody": {
1026             "content": {
1027                 "application/json": {
1028                     "schema": {
1029                         "$ref": "#/components/schemas/
1030                             DashboardWindowUpdatedDTO"
1031                     }
1032                 },
1033                 "required": true
1034             },
1035             "responses": {
1036                 "200": {
1037                     "description": "OK"
1038                 }
1039             }
1040         }
1041     }
1042 }
```

```

1040     }
1041 },
1042 "/dashboards/widgets/{widgetId}/config": {
1043   "patch": {
1044     "tags": [
1045       "CloudSherpa Dashboard"
1046     ],
1047     "summary": "Update a widget's visual or data
1048               configuration",
1049     "operationId": "updateWidgetConfig",
1050     "parameters": [
1051       {
1052         "name": "widgetId",
1053         "in": "path",
1054         "required": true,
1055         "schema": {
1056           "type": "string",
1057           "format": "uuid"
1058         }
1059       },
1060     ],
1061     "requestBody": {
1062       "content": {
1063         "application/json": {
1064           "schema": {
1065             "oneOf": [
1066               {
1067                 "$ref": "#/components/schemas/
1068                   ChartWidgetConfigUpdatedDTO"
1069               },
1070               {
1071                 "$ref": "#/components/schemas/
1072                   KpiWidgetConfigUpdatedDTO"
1073               }
1074             ]
1075           }
1076         },
1077         "required": true
1078       },
1079       "responses": {
1080         "200": {
1081           "description": "OK"
1082         }
1083       }
1084     }
1085   }
1086 }

```

```

1081     }
1082   }
1083 },
1084 "/aws/resources/{resourceId}/status": {
1085   "patch": {
1086     "tags": [
1087       "aws-connection-persistence-controller"
1088     ],
1089     "operationId": "setResourceStatus",
1090     "parameters": [
1091       {
1092         "name": "resourceId",
1093         "in": "path",
1094         "required": true,
1095         "schema": {
1096           "type": "string",
1097           "format": "uuid"
1098         }
1099       }
1100     ],
1101     "requestBody": {
1102       "content": {
1103         "application/json": {
1104           "schema": {
1105             "$ref": "#/components/schemas/UpdateResourceStatusRequest"
1106           }
1107         }
1108       },
1109       "required": true
1110     },
1111     "responses": {
1112       "200": {
1113         "description": "OK"
1114       }
1115     }
1116   },
1117 },
1118 "/aws/connections/{accountId}/name": {
1119   "patch": {
1120     "tags": [
1121       "aws-connection-persistence-controller"
1122     ],
1123     "operationId": "setAccountDisplayName",

```

```
1124     "parameters": [  
1125         {  
1126             "name": "accountId",  
1127             "in": "path",  
1128             "required": true,  
1129             "schema": {  
1130                 "type": "string",  
1131                 "format": "uuid"  
1132             }  
1133         }  
1134     ],  
1135     "requestBody": {  
1136         "content": {  
1137             "application/json": {  
1138                 "schema": {  
1139                     "$ref": "#/components/schemas/  
                        UpdateAccountNameRequest"  
1140                 }  
1141             }  
1142         },  
1143         "required": true  
1144     },  
1145     "responses": {  
1146         "200": {  
1147             "description": "OK"  
1148         }  
1149     }  
1150 },  
1151 },  
1152 "/stream": {  
1153     "get": {  
1154         "tags": [  
1155             "sse-controller"  
1156         ],  
1157         "operationId": "subscribe",  
1158         "responses": {  
1159             "200": {  
1160                 "description": "OK",  
1161                 "content": {  
1162                     "text/event-stream": {  
1163                         "schema": {  
1164                             "$ref": "#/components/schemas/SseEmitter"  
1165                         }  
1166                     }  
1167                 }  
1168             }  
1169         }  
1170     }  
1171 }
```

```
1167     }
1168   }
1169 }
1170 }
1171 },
1172 "/optimization/recommendations": {
1173   "get": {
1174     "tags": [
1175       "Optimization Recommendations"
1176     ],
1177     "summary": "List optimization recommendations",
1178     "description": "Returns optimization recommendations
1179       matching the supplied filters.",
1179     "operationId": "getRecommendations",
1180     "parameters": [
1181       {
1182         "name": "status",
1183         "in": "query",
1184         "description": "Recommendation lifecycle status",
1185         "required": false,
1186         "schema": {
1187           "type": "string"
1188         },
1189         "example": "ACTIVE"
1190       },
1191       {
1192         "name": "provider",
1193         "in": "query",
1194         "description": "Cloud provider",
1195         "required": false,
1196         "schema": {
1197           "type": "string"
1198         },
1199         "example": "AWS"
1200       },
1201       {
1202         "name": "resourceType",
1203         "in": "query",
1204         "description": "Cloud resource type",
1205         "required": false,
1206         "schema": {
1207           "type": "string"
1208         },
1209         "example": "compute_instance"
```

```

1210     },
1211     {
1212         "name": "actionType",
1213         "in": "query",
1214         "description": "Recommended action type",
1215         "required": false,
1216         "schema": {
1217             "type": "string"
1218         },
1219         "example": "DOWNSIZE"
1220     },
1221     {
1222         "name": "resourceId",
1223         "in": "query",
1224         "description": "Resource UUID",
1225         "required": false,
1226         "schema": {
1227             "type": "string",
1228             "format": "uuid"
1229         }
1230     }
1231 ],
1232 "responses": {
1233     "200": {
1234         "description": "OK",
1235         "content": {
1236             "*/*": {
1237                 "schema": {
1238                     "type": "array",
1239                     "items": {
1240                         "type": "object",
1241                         "additionalProperties": {}
1242                     }
1243                 }
1244             }
1245         }
1246     }
1247 },
1248 },
1249 },
1250 "/optimization/recommendations/{recommendationId}": {
1251     "get": {
1252         "tags": [
1253             "Optimization Recommendations"

```

```

1254     ],
1255     "summary": "Get an optimization recommendation",
1256     "description": "Returns a single optimization
1257                     recommendation by its identifier.",
1258     "operationId": "getRecommendation",
1259     "parameters": [
1260         {
1261             "name": "recommendationId",
1262             "in": "path",
1263             "description": "Recommendation UUID",
1264             "required": true,
1265             "schema": {
1266                 "type": "string",
1267                 "format": "uuid"
1268             }
1269         ],
1270     "responses": {
1271         "200": {
1272             "description": "OK",
1273             "content": {
1274                 "*/*": {
1275                     "schema": {
1276                         "type": "object",
1277                         "additionalProperties": {}
1278                     }
1279                 }
1280             }
1281         }
1282     }
1283 },
1284 "/optimization/recommendations/summary": {
1285     "get": {
1286         "tags": [
1287             "Optimization Recommendations"
1288         ],
1289         "summary": "Get recommendation summary",
1290         "description": "Returns recommendation counts grouped by
1291                       status and action type.",
1292         "operationId": "getRecommendationSummary",
1293         "responses": {
1294             "200": {
1295                 "description": "OK",

```

```
1296         "content": {
1297             "*/*": {
1298                 "schema": {
1299                     "type": "object",
1300                     "additionalProperties": {}
1301                 }
1302             }
1303         }
1304     }
1305 }
1306 },
1307 "/mock": {
1308     "get": {
1309         "tags": [
1310             "mock-controller"
1311         ],
1312         "operationId": "getMethodNames",
1313         "responses": {
1314             "200": {
1315                 "description": "OK",
1316                 "content": {
1317                     "*/*": {
1318                         "schema": {
1319                             "type": "string"
1320                         }
1321                     }
1322                 }
1323             }
1324         }
1325     }
1326 },
1327 "/intelligence/mock/usage-sample": {
1328     "get": {
1329         "tags": [
1330             "intelligence-mock-controller"
1331         ],
1332         "operationId": "sampleUsage",
1333         "responses": {
1334             "200": {
1335                 "description": "OK",
1336                 "content": {
1337                     "*/*": {
1338                         "schema": {
```

```
1340         "type": "array",
1341         "items": {
1342             "$ref": "#/components/schemas/
                  TimestampedNumericDataPoint "
1343         }
1344     }
1345 }
1346 }
1347 }
1348 }
1349 }
1350 },
1351 "/billing/charges": {
1352     "get": {
1353         "tags": [
1354             "Billing"
1355         ],
1356         "summary": "Get billing resources",
1357         "operationId": "getResources_1",
1358         "responses": {
1359             "200": {
1360                 "description": "Resources returned successfully",
1361                 "content": {
1362                     "*/*": {
1363                         "schema": {
1364                             "type": "array",
1365                             "items": {
1366                                 "$ref": "#/components/schemas/
                                      BillingChargeResponse "
1367                             }
1368                         }
1369                     }
1370                 }
1371             }
1372         }
1373     }
1374 },
1375 "/azure/accounts/{accountId}/resources/count": {
1376     "get": {
1377         "tags": [
1378             "connection-query-controller"
1379         ],
1380         "operationId": "getResourceCount",
1381         "parameters": [
```

```
1382     {
1383         "name": "accountId",
1384         "in": "path",
1385         "required": true,
1386         "schema": {
1387             "type": "string",
1388             "format": "uuid"
1389         }
1390     }
1391 ],
1392 "responses": {
1393     "200": {
1394         "description": "OK",
1395         "content": {
1396             "*/*": {
1397                 "schema": {
1398                     "$ref": "#/components/schemas/
ResourceCountResponse"
1399                 }
1400             }
1401         }
1402     }
1403 }
1404 },
1405 "/gcp/accounts/{accountId}/resources/count": {
1406     "get": {
1407         "tags": [
1408             "connection-query-controller"
1409         ],
1410         "operationId": "getResourceCount_1",
1411         "parameters": [
1412             {
1413                 "name": "accountId",
1414                 "in": "path",
1415                 "required": true,
1416                 "schema": {
1417                     "type": "string",
1418                     "format": "uuid"
1419                 }
1420             }
1421         ],
1422         "responses": {
1423             "200": {
```

```
1425         "description": "OK",
1426         "content": {
1427             "*/*": {
1428                 "schema": {
1429                     "$ref": "#/components/schemas/
                        ResourceCountResponse"
1430                 }
1431             }
1432         }
1433     }
1434 }
1435 },
1436 "/aws/accounts/{accountId}/resources/count": {
1437     "get": {
1438         "tags": [
1439             "connection-query-controller"
1440         ],
1441         "operationId": "getResourceCount_2",
1442         "parameters": [
1443             {
1444                 "name": "accountId",
1445                 "in": "path",
1446                 "required": true,
1447                 "schema": {
1448                     "type": "string",
1449                     "format": "uuid"
1450                 }
1451             }
1452         ],
1453         "responses": {
1454             "200": {
1455                 "description": "OK",
1456                 "content": {
1457                     "*/*": {
1458                         "schema": {
1459                             "$ref": "#/components/schemas/
                        ResourceCountResponse"
1460                         }
1461                     }
1462                 }
1463             }
1464         }
1465     }
1466 }
```

```

1467 },
1468 "/aws/accounts/{accountId}/resources": {
1469   "get": {
1470     "tags": [
1471       "connection-query-controller"
1472     ],
1473     "operationId": "getResourcesForAccount",
1474     "parameters": [
1475       {
1476         "name": "accountId",
1477         "in": "path",
1478         "required": true,
1479         "schema": {
1480           "type": "string",
1481           "format": "uuid"
1482         }
1483       }
1484     ],
1485     "responses": {
1486       "200": {
1487         "description": "OK",
1488         "content": {
1489           "*/*": {
1490             "schema": {
1491               "type": "array",
1492               "items": {
1493                 "$ref": "#/components/schemas/Resource"
1494               }
1495             }
1496           }
1497         }
1498       }
1499     }
1500   }
1501 },
1502 "/gcp/accounts/{accountId}/resources": {
1503   "get": {
1504     "tags": [
1505       "connection-query-controller"
1506     ],
1507     "operationId": "getResourcesForAccount_1",
1508     "parameters": [
1509       {
1510         "name": "accountId",

```

```
1511         "in": "path",
1512         "required": true,
1513         "schema": {
1514             "type": "string",
1515             "format": "uuid"
1516         }
1517     },
1518 ],
1519 "responses": {
1520     "200": {
1521         "description": "OK",
1522         "content": {
1523             "*/*": {
1524                 "schema": {
1525                     "type": "array",
1526                     "items": {
1527                         "$ref": "#/components/schemas/Resource"
1528                     }
1529                 }
1530             }
1531         }
1532     }
1533 },
1534 },
1535 },
1536 "/azure/accounts/{accountId}/resources": {
1537     "get": {
1538         "tags": [
1539             "connection-query-controller"
1540         ],
1541         "operationId": "getResourcesForAccount_2",
1542         "parameters": [
1543             {
1544                 "name": "accountId",
1545                 "in": "path",
1546                 "required": true,
1547                 "schema": {
1548                     "type": "string",
1549                     "format": "uuid"
1550                 }
1551             }
1552         ],
1553         "responses": {
1554             "200": {
```

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1555         "description": "OK",
1556         "content": {
1557             "*/*": {
1558                 "schema": {
1559                     "type": "array",
1560                     "items": {
1561                         "$ref": "#/components/schemas/Resource"
1562                     }
1563                 }
1564             }
1565         }
1566     },
1567     "/azure/accounts/{accountId}": {
1568         "get": {
1569             "tags": [
1570                 "connection-query-controller"
1571             ],
1572             "operationId": "getAccount",
1573             "parameters": [
1574                 {
1575                     "name": "accountId",
1576                     "in": "path",
1577                     "required": true,
1578                     "schema": {
1579                         "type": "string",
1580                         "format": "uuid"
1581                     }
1582                 }
1583             ],
1584             "responses": {
1585                 "200": {
1586                     "description": "OK",
1587                     "content": {
1588                         "*/*": {
1589                             "schema": {
1590                                 "$ref": "#/components/schemas/CloudAccountDetailsResponse"
1591                             }
1592                         }
1593                     }
1594                 }
1595             }
1596         }
1597     }

```

```
1598     }
1599   }
1600 },
1601 "/gcp/accounts/{accountId}": {
1602   "get": {
1603     "tags": [
1604       "connection-query-controller"
1605     ],
1606     "operationId": "getAccount_1",
1607     "parameters": [
1608       {
1609         "name": "accountId",
1610         "in": "path",
1611         "required": true,
1612         "schema": {
1613           "type": "string",
1614           "format": "uuid"
1615         }
1616       }
1617     ],
1618     "responses": {
1619       "200": {
1620         "description": "OK",
1621         "content": {
1622           "*/*": {
1623             "schema": {
1624               "$ref": "#/components/schemas/CloudAccountDetailsResponse"
1625             }
1626           }
1627         }
1628       }
1629     }
1630   }
1631 },
1632 "/aws/accounts/{accountId}": {
1633   "get": {
1634     "tags": [
1635       "connection-query-controller"
1636     ],
1637     "operationId": "getAccount_2",
1638     "parameters": [
1639       {
1640         "name": "accountId",
```

```
1641         "in": "path",
1642         "required": true,
1643         "schema": {
1644             "type": "string",
1645             "format": "uuid"
1646         }
1647     },
1648 ],
1649 "responses": {
1650     "200": {
1651         "description": "OK",
1652         "content": {
1653             "*/*": {
1654                 "schema": {
1655                     "$ref": "#/components/schemas/
                        CloudAccountDetailsResponse"
1656                 }
1657             }
1658         }
1659     }
1660 }
1661 },
1662 },
1663 "/auth/me": {
1664     "get": {
1665         "tags": [
1666             "Authentication"
1667         ],
1668         "summary": "Get current authenticated user",
1669         "operationId": "me",
1670         "responses": {
1671             "200": {
1672                 "description": "Current authenticated user",
1673                 "content": {
1674                     "*/*": {
1675                         "schema": {
1676                             "$ref": "#/components/schemas/AuthUserResponse"
1677                         }
1678                     }
1679                 }
1680             },
1681             "401": {
1682                 "description": "Unauthenticated",
1683                 "content": {
```

```

1684         "*/*": {
1685             "schema": {
1686                 "$ref": "#/components/schemas/AuthUserResponse"
1687             }
1688         }
1689     }
1690 }
1691 }
1692 },
1693 "/analytics/resource-names": {
1694     "get": {
1695         "tags": [
1696             "CloudSherpa Analytics"
1697         ],
1698         "operationId": "getResourceNames",
1699         "responses": {
1700             "200": {
1701                 "description": "OK",
1702                 "content": {
1703                     "*/*": {
1704                         "schema": {
1705                             "type": "array",
1706                             "items": {
1707                                 "$ref": "#/components/schemas/ResourceNameDto"
1708                             }
1709                         }
1710                     }
1711                 }
1712             }
1713         }
1714     }
1715 },
1716 "/analytics/resource-metrics": {
1717     "get": {
1718         "tags": [
1719             "CloudSherpa Analytics"
1720         ],
1721         "summary": "For each resource return available/recorded metrics",
1722         "operationId": "getResourceMetrics",
1723         "responses": {
1724             "200": {

```

```

1726         "description": "Succesfully fetched available metrics
1727             for each resource",
1728         "content": {
1729             "*/*": {
1730                 "schema": {
1731                     "type": "array",
1732                     "items": {
1733                         "$ref": "#/components/schemas/
1734                             ResourceMetricsGroupDto"
1735                     }
1736                 }
1737             }
1738         },
1739     },
1740     "/analytics/historical": {
1741         "get": {
1742             "tags": [
1743                 "CloudSherpa Analytics"
1744             ],
1745             "summary": "Get authoratative metric data",
1746             "operationId": "getHistoricalData",
1747             "parameters": [
1748                 {
1749                     "name": "from",
1750                     "in": "query",
1751                     "required": true,
1752                     "schema": {
1753                         "type": "string"
1754                     }
1755                 },
1756                 {
1757                     "name": "to",
1758                     "in": "query",
1759                     "required": true,
1760                     "schema": {
1761                         "type": "string"
1762                     }
1763                 },
1764                 {
1765                     "name": "interval",
1766                     "in": "query",

```

```

1768         "required": false,
1769         "schema": {
1770             "type": "string",
1771             "default": "daily"
1772         }
1773     },
1774 ],
1775 "responses": {
1776     "204": {
1777         "description": "No metrics found for selected time
1778             window",
1779         "content": {
1780             "*/*": {
1781                 "schema": {
1782                     "type": "array",
1783                     "items": {
1784                         "$ref": "#/components/schemas/MetricDto"
1785                     }
1786                 }
1787             }
1788         },
1789         "200": {
1790             "description": "Return metrics found",
1791             "content": {
1792                 "*/*": {
1793                     "schema": {
1794                         "type": "array",
1795                         "items": {
1796                             "$ref": "#/components/schemas/
1797                                 AggregatedMetric"
1798                         }
1799                     }
1800                 }
1801             }
1802         }
1803     }
1804 },
1805 "/dashboards/{dashboardId}": {
1806     "delete": {
1807         "tags": [
1808             "CloudSherpa Dashboard"
1809         ],

```

```
1810     "summary": "Delete a dashboard",
1811     "operationId": "deleteDashboard",
1812     "parameters": [
1813       {
1814         "name": "dashboardId",
1815         "in": "path",
1816         "required": true,
1817         "schema": {
1818           "type": "string",
1819           "format": "uuid"
1820         }
1821       }
1822     ],
1823     "responses": {
1824       "204": {
1825         "description": "No Content"
1826       }
1827     }
1828   },
1829 },
1830 "/dashboards/widgets/{widgetId}": {
1831   "delete": {
1832     "tags": [
1833       "CloudSherpa Dashboard"
1834     ],
1835     "summary": "Delete a widget from a dashboard",
1836     "operationId": "deleteWidget",
1837     "parameters": [
1838       {
1839         "name": "widgetId",
1840         "in": "path",
1841         "required": true,
1842         "schema": {
1843           "type": "string",
1844           "format": "uuid"
1845         }
1846       }
1847     ],
1848     "responses": {
1849       "204": {
1850         "description": "No Content"
1851       }
1852     }
1853   }
1854 }
```

```
1854 },
1855 "/aws/connections/{accountId}": {
1856   "delete": {
1857     "tags": [
1858       "aws-connection-persistence-controller"
1859     ],
1860     "operationId": "deleteAccount",
1861     "parameters": [
1862       {
1863         "name": "accountId",
1864         "in": "path",
1865         "required": true,
1866         "schema": {
1867           "type": "string",
1868           "format": "uuid"
1869         }
1870       }
1871     ],
1872     "responses": {
1873       "200": {
1874         "description": "OK"
1875       }
1876     }
1877   }
1878 },
1879 },
1880 "components": {
1881   "schemas": {
1882     "WidgetLayoutUpdatedDTO": {
1883       "type": "object",
1884       "properties": {
1885         "id": {
1886           "type": "string",
1887           "format": "uuid"
1888         },
1889         "x": {
1890           "type": "integer",
1891           "format": "int32"
1892         },
1893         "y": {
1894           "type": "integer",
1895           "format": "int32"
1896         },
1897         "w": {
```

```
1898         "type": "integer",
1899         "format": "int32"
1900     },
1901     "h": {
1902         "type": "integer",
1903         "format": "int32"
1904     }
1905 },
1906 "ThemeUpdateRequest": {
1907     "type": "object",
1908     "properties": {
1909         "theme": {
1910             "type": "string"
1911         }
1912     }
1913 },
1914 "ResourceUsageForecastRequestDto": {
1915     "type": "object",
1916     "properties": {
1917         "resourceId": {
1918             "type": "string",
1919             "format": "uuid"
1920         },
1921         "metricType": {
1922             "type": "string"
1923         },
1924         "forecastHorizon": {
1925             "type": "string",
1926             "format": "date-time"
1927         }
1928     }
1929 },
1930 "ResourceUsageForecastResponseDto": {
1931     "type": "object",
1932     "properties": {
1933         "horizonTimestamps": {
1934             "type": "array",
1935             "items": {
1936                 "type": "string",
1937                 "format": "date-time"
1938             }
1939         },
1940         "predictedValues": {
```

```
1942         "type": "array",
1943         "items": {
1944             "type": "number"
1945         }
1946     },
1947     "q1Values": {
1948         "type": "array",
1949         "items": {
1950             "type": "number"
1951         }
1952     },
1953     "q3Values": {
1954         "type": "array",
1955         "items": {
1956             "type": "number"
1957         }
1958     }
1959 },
1960 "BillingForecastRequest": {
1961     "type": "object",
1962     "properties": {
1963         "forecastSteps": {
1964             "type": "integer",
1965             "format": "int32"
1966         }
1967     }
1968 },
1969 },
1970 "BillingForecastResponseDto": {
1971     "type": "object",
1972     "properties": {
1973         "cumalativeBillingForecastValue": {
1974             "type": "number"
1975         },
1976         "cumalitivePastForecastingValue": {
1977             "type": "number"
1978         },
1979         "billingForecastSeries": {
1980             "type": "object",
1981             "additionalProperties": {
1982                 "$ref": "#/components/schemas/BillingForecastValue"
1983             }
1984         },
1985         "failedForecastCharges": {
```

```
1986         "type": "array",
1987         "items": {
1988             "type": "string"
1989         }
1990     },
1991     "pastVariance": {
1992         "type": "number"
1993     },
1994     "dailyBurnRate": {
1995         "type": "number"
1996     },
1997     "highestCostDriver": {
1998         "type": "string"
1999     },
2000     "highestCostAcceleration": {
2001         "type": "string"
2002     }
2003 }
2004 },
2005 "BillingForecastValue": {
2006     "type": "object",
2007     "properties": {
2008         "value": {
2009             "type": "number"
2010         },
2011         "percentageOfTotal": {
2012             "type": "number"
2013         },
2014         "chargeLabel": {
2015             "type": "string"
2016         }
2017     }
2018 },
2019 "BillingForecastIndividualChargesRequestDto": {
2020     "type": "object",
2021     "properties": {
2022         "chargeIds": {
2023             "type": "array",
2024             "items": {
2025                 "type": "string"
2026             }
2027         },
2028         "forecastSteps": {
2029             "type": "integer",
```

```
2030         "format": "int32"
2031     }
2032 }
2033 },
2034 "GcpBillingConfigDto": {
2035     "type": "object",
2036     "properties": {
2037         "billingId": {
2038             "type": "string"
2039         },
2040         "dataset": {
2041             "type": "string"
2042         }
2043     }
2044 },
2045 "GcpCredentialsDto": {
2046     "type": "object",
2047     "properties": {
2048         "projectId": {
2049             "type": "string"
2050         },
2051         "serviceAccountJson": {
2052             "type": "string"
2053         }
2054     }
2055 },
2056 "PersistGcpConnectionRequest": {
2057     "type": "object",
2058     "properties": {
2059         "userId": {
2060             "type": "string",
2061             "format": "uuid"
2062         },
2063         "accountId": {
2064             "type": "string"
2065         },
2066         "displayName": {
2067             "type": "string"
2068         },
2069         "ingestionPeriod": {
2070             "type": "integer",
2071             "format": "int32"
2072         },
2073         "credentials": {
```

```
2074         "$ref": "#/components/schemas/GcpCredentialsDto"
2075     },
2076     "resources": {
2077         "type": "array",
2078         "items": {
2079             "$ref": "#/components/schemas/ResourceSelectionDto"
2080         }
2081     },
2082     "billingConfig": {
2083         "$ref": "#/components/schemas/GcpBillingConfigDto"
2084     }
2085 }
2086 },
2087 "ResourceSelectionDto": {
2088     "type": "object",
2089     "properties": {
2090         "resourceId": {
2091             "type": "string"
2092         },
2093         "serviceType": {
2094             "type": "string"
2095         },
2096         "resourceType": {
2097             "type": "string"
2098         },
2099         "resourceName": {
2100             "type": "string"
2101         },
2102         "region": {
2103             "type": "string"
2104         },
2105         "tags": {
2106             "type": "object",
2107             "additionalProperties": {}
2108         },
2109         "active": {
2110             "type": "boolean"
2111         }
2112     }
2113 },
2114 "DashboardCreateDTO": {
2115     "type": "object",
2116     "properties": {
2117         "userId": {
```

```
2118         "type": "string",
2119         "format": "uuid"
2120     },
2121     "id": {
2122         "type": "string",
2123         "format": "uuid"
2124     },
2125     "displayName": {
2126         "type": "string"
2127     }
2128 },
2129 },
2130 "ChartWidgetDTO": {
2131     "allOf": [
2132         {
2133             "$ref": "#/components/schemas/WidgetDTO"
2134         },
2135         {
2136             "type": "object",
2137             "properties": {
2138                 "id": {
2139                     "type": "string",
2140                     "format": "uuid"
2141                 },
2142                 "widgetType": {
2143                     "type": "string",
2144                     "enum": [
2145                         "KPI",
2146                         "CHART"
2147                     ]
2148                 },
2149                 "displayName": {
2150                     "type": "string"
2151                 },
2152                 "startX": {
2153                     "type": "integer",
2154                     "format": "int32"
2155                 },
2156                 "startY": {
2157                     "type": "integer",
2158                     "format": "int32"
2159                 },
2160                 "width": {
2161                     "type": "integer",
```

```
2162         "format": "int32"
2163     },
2164     "height": {
2165         "type": "integer",
2166         "format": "int32"
2167     },
2168     "chartType": {
2169         "type": "string",
2170         "enum": [
2171             "gauge_chart",
2172             "line_chart"
2173         ]
2174     },
2175     "provider": {
2176         "type": "string",
2177         "enum": [
2178             "AWS",
2179             "AZURE",
2180             "GCP"
2181         ]
2182     },
2183     "accountId": {
2184         "type": "string",
2185         "format": "uuid"
2186     },
2187     "resourceId": {
2188         "type": "string",
2189         "format": "uuid"
2190     },
2191     "metricType": {
2192         "type": "string"
2193     },
2194     "metricName": {
2195         "type": "string"
2196     }
2197 }
2198 }
2199 ]
2200 },
2201 "DashboardDTO": {
2202     "type": "object",
2203     "properties": {
2204         "id": {
2205             "type": "string",
```

```
2206         "format": "uuid"
2207     },
2208     "displayName": {
2209         "type": "string"
2210     },
2211     "timeFrom": {
2212         "type": "string",
2213         "format": "date-time"
2214     },
2215     "timeTo": {
2216         "type": "string",
2217         "format": "date-time"
2218     },
2219     "predefinedTime": {
2220         "type": "string",
2221         "enum": [
2222             "T_5_MIN",
2223             "T_15_MIN",
2224             "T_30_MIN",
2225             "T_1_HOUR",
2226             "T_6_HOUR",
2227             "T_12_HOUR",
2228             "T_24_HOUR",
2229             "T_7_DAYS",
2230             "T_30_DAYS"
2231         ]
2232     },
2233     "current": {
2234         "type": "boolean"
2235     },
2236     "widgets": {
2237         "type": "array",
2238         "items": {
2239             "oneOf": [
2240                 {
2241                     "$ref": "#/components/schemas/ChartWidgetDTO"
2242                 },
2243                 {
2244                     "$ref": "#/components/schemas/KpiWidgetDTO"
2245                 }
2246             ]
2247         }
2248     }
2249 }
```

```
2250     },
2251     "KpiWidgetDTO": {
2252       "allOf": [
2253         {
2254           "$ref": "#/components/schemas/WidgetDTO"
2255         },
2256         {
2257           "type": "object",
2258           "properties": {
2259             "id": {
2260               "type": "string",
2261               "format": "uuid"
2262             },
2263             "widgetType": {
2264               "type": "string",
2265               "enum": [
2266                 "KPI",
2267                 "CHART"
2268               ]
2269             },
2270             "displayName": {
2271               "type": "string"
2272             },
2273             "startX": {
2274               "type": "integer",
2275               "format": "int32"
2276             },
2277             "startY": {
2278               "type": "integer",
2279               "format": "int32"
2280             },
2281             "width": {
2282               "type": "integer",
2283               "format": "int32"
2284             },
2285             "height": {
2286               "type": "integer",
2287               "format": "int32"
2288             },
2289             "chargeIds": {
2290               "type": "array",
2291               "items": {
2292                 "type": "string"
2293               }
2294             }
2295           }
2296         }
2297       ]
2298     }
2299   }
2300 }
```

```
2294         },
2295         "aggregationWindowDays": {
2296             "type": "integer",
2297             "format": "int32"
2298         }
2299     }
2300 }
2301 ]
2302 },
2303 "WidgetDTO": {
2304     "discriminator": {
2305         "propertyName": "widgetType"
2306     },
2307     "properties": {
2308         "widgetType": {
2309             "type": "string"
2310         }
2311     },
2312     "required": [
2313         "widgetType"
2314     ]
2315 },
2316 "BillingKpiRequest": {
2317     "type": "object",
2318     "properties": {
2319         "chargeIds": {
2320             "type": "array",
2321             "items": {
2322                 "type": "string"
2323             }
2324         },
2325         "from": {
2326             "type": "string"
2327         },
2328         "to": {
2329             "type": "string"
2330         },
2331         "aggregation": {
2332             "type": "string"
2333         }
2334     }
2335 },
2336 "BillingKpiResponse": {
2337     "type": "object",
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2338     "properties": {
2339       "value": {
2340         "type": "number"
2341       },
2342       "currency": {
2343         "type": "string"
2344       },
2345       "selectedChargeCount": {
2346         "type": "integer",
2347         "format": "int32"
2348       },
2349       "timeLabel": {
2350         "type": "string"
2351       },
2352       "updatedAt": {
2353         "type": "string"
2354       },
2355       "previousValue": {
2356         "type": "number"
2357       }
2358     },
2359     "AzureCredentialsDto": {
2360       "type": "object",
2361       "properties": {
2362         "subscriptionId": {
2363           "type": "string"
2364         },
2365         "tenantId": {
2366           "type": "string"
2367         },
2368         "clientId": {
2369           "type": "string"
2370         },
2371         "clientSecret": {
2372           "type": "string"
2373         }
2374       }
2375     },
2376     "PersistAzureConnectionRequest": {
2377       "type": "object",
2378       "properties": {
2379         "userId": {
2380           "type": "string",
2381
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2382         "format": "uuid"
2383     },
2384     "accountId": {
2385         "type": "string"
2386     },
2387     "displayName": {
2388         "type": "string"
2389     },
2390     "ingestionPeriod": {
2391         "type": "integer",
2392         "format": "int32"
2393     },
2394     "credentials": {
2395         "$ref": "#/components/schemas/AzureCredentialsDto"
2396     },
2397     "resources": {
2398         "type": "array",
2399         "items": {
2400             "$ref": "#/components/schemas/ResourceSelectionDto"
2401         }
2402     }
2403 },
2404 },
2405 "AwsCredentialsDto": {
2406     "type": "object",
2407     "properties": {
2408         "accessKeyId": {
2409             "type": "string"
2410         },
2411         "secretAccessKey": {
2412             "type": "string"
2413         }
2414     }
2415 },
2416 "BillingConfigDto": {
2417     "type": "object",
2418     "properties": {
2419         "bucketName": {
2420             "type": "string"
2421         },
2422         "bucketRegion": {
2423             "type": "string"
2424         },
2425         "exportPrefix": {
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2426         "type": "string"
2427     },
2428     "exportName": {
2429         "type": "string"
2430     }
2431 }
2432 },
2433 "PersistAwsConnectionRequest": {
2434     "type": "object",
2435     "properties": {
2436         "userId": {
2437             "type": "string",
2438             "format": "uuid"
2439         },
2440         "accountId": {
2441             "type": "string"
2442         },
2443         "displayName": {
2444             "type": "string"
2445         },
2446         "ingestionPeriod": {
2447             "type": "integer",
2448             "format": "int32"
2449         },
2450         "credentials": {
2451             "$ref": "#/components/schemas/AwsCredentialsDto"
2452         },
2453         "resources": {
2454             "type": "array",
2455             "items": {
2456                 "$ref": "#/components/schemas/ResourceSelectionDto"
2457             }
2458         },
2459         "billingConfig": {
2460             "$ref": "#/components/schemas/BillingConfigDto"
2461         }
2462     }
2463 },
2464 "RegisterRequest": {
2465     "type": "object",
2466     "properties": {
2467         "email": {
2468             "type": "string"
2469         },
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2470         "username": {
2471             "type": "string"
2472         },
2473         "password": {
2474             "type": "string"
2475         }
2476     },
2477 },
2478 "LoginRequest": {
2479     "type": "object",
2480     "properties": {
2481         "email": {
2482             "type": "string"
2483         },
2484         "password": {
2485             "type": "string"
2486         }
2487     }
2488 },
2489 "AuthUserResponse": {
2490     "type": "object",
2491     "properties": {
2492         "userId": {
2493             "type": "string",
2494             "format": "uuid"
2495         },
2496         "email": {
2497             "type": "string"
2498         },
2499         "username": {
2500             "type": "string"
2501         },
2502         "token": {
2503             "type": "string"
2504         }
2505     }
2506 },
2507 "CloudCredentialsDto": {
2508     "type": "object",
2509     "properties": {
2510         "accessKeyId": {
2511             "type": "string"
2512         },
2513         "secretAccessKey": {
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2514         "type": "string"
2515     },
2516     "awsRegion": {
2517         "type": "string"
2518     },
2519     "subscriptionId": {
2520         "type": "string"
2521     },
2522     "tenantId": {
2523         "type": "string"
2524     },
2525     "clientId": {
2526         "type": "string"
2527     },
2528     "clientSecret": {
2529         "type": "string"
2530     },
2531     "projectId": {
2532         "type": "string"
2533     },
2534     "serviceAccountJson": {
2535         "type": "string"
2536     }
2537 },
2538 "ResourceDiscoveryDto": {
2539     "type": "object",
2540     "properties": {
2541         "services": {
2542             "type": "array",
2543             "items": {
2544                 "type": "string"
2545             }
2546         },
2547         "credentials": {
2548             "$ref": "#/components/schemas/CloudCredentialsDto"
2549         }
2550     }
2551 },
2552 "ResourceDetailDto": {
2553     "type": "object",
2554     "properties": {
2555         "resourceId": {
2556             "type": "string"
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2558     },
2559     "name": {
2560         "type": "string"
2561     },
2562     "resourceType": {
2563         "type": "string"
2564     },
2565     "serviceCategory": {
2566         "type": "string"
2567     },
2568     "region": {
2569         "type": "string"
2570     },
2571     "tags": {
2572         "type": "object",
2573         "additionalProperties": {
2574             "type": "string"
2575         }
2576     }
2577 },
2578 "ResourceMetricHistoricalRequestDto": {
2579     "type": "object",
2580     "properties": {
2581         "resourceId": {
2582             "type": "string",
2583             "format": "uuid"
2584         },
2585         "metricType": {
2586             "type": "string"
2587         },
2588         "fromDate": {
2589             "type": "string",
2590             "format": "date-time"
2591         }
2592     }
2593 },
2594 "ResourceMetricHistoricalResponseDto": {
2595     "type": "object",
2596     "properties": {
2597         "values": {
2598             "type": "array",
2599             "items": {
2600                 "type": "number"
2601             }
2602         }
2603     }
2604 }
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2602     }
2603   },
2604   "timestamps": {
2605     "type": "array",
2606     "items": {
2607       "type": "string",
2608       "format": "date-time"
2609     }
2610   }
2611 },
2612 "DownsampledSeriesRequestDto": {
2613   "type": "object",
2614   "properties": {
2615     "resourceId": {
2616       "type": "string",
2617       "format": "uuid"
2618     },
2619     "metricName": {
2620       "type": "string"
2621     },
2622     "from": {
2623       "type": "string",
2624       "format": "date-time"
2625     },
2626     "to": {
2627       "type": "string",
2628       "format": "date-time"
2629     }
2630   }
2631 },
2632 "NormalizedMetrics": {
2633   "type": "object",
2634   "properties": {
2635     "metricId": {
2636       "type": "string",
2637       "format": "uuid"
2638     },
2639     "resourceId": {
2640       "type": "string",
2641       "format": "uuid"
2642     },
2643     "recordedAt": {
2644       "type": "string",
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2646         "format": "date-time"
2647     },
2648     "metricType": {
2649         "type": "string"
2650     },
2651     "metricName": {
2652         "type": "string"
2653     },
2654     "metricValue": {
2655         "type": "number"
2656     },
2657     "unit": {
2658         "type": "string"
2659     },
2660     "currency": {
2661         "type": "string"
2662     },
2663     "periodStart": {
2664         "type": "string",
2665         "format": "date-time"
2666     },
2667     "periodEnd": {
2668         "type": "string",
2669         "format": "date-time"
2670     }
2671 },
2672 "DashboardWindowUpdatedDTO": {
2673     "type": "object",
2674     "properties": {
2675         "newTime": {
2676             "type": "string",
2677             "enum": [
2678                 "T_5_MIN",
2679                 "T_15_MIN",
2680                 "T_30_MIN",
2681                 "T_1_HOUR",
2682                 "T_6_HOUR",
2683                 "T_12_HOUR",
2684                 "T_24_HOUR",
2685                 "T_7_DAYS",
2686                 "T_30_DAYS"
2687             ]
2688         }
2689     }
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2690     }
2691 },
2692 "ChartWidgetConfigUpdatedDTO": {
2693   "allOf": [
2694     {
2695       "$ref": "#/components/schemas/WidgetConfigUpdatedDTO"
2696     },
2697     {
2698       "type": "object",
2699       "properties": {
2700         "id": {
2701           "type": "string",
2702           "format": "uuid"
2703         },
2704         "widgetType": {
2705           "type": "string",
2706           "enum": [
2707             "KPI",
2708             "CHART"
2709           ]
2710         },
2711         "displayName": {
2712           "type": "string"
2713         },
2714         "chartType": {
2715           "type": "string",
2716           "enum": [
2717             "gauge_chart",
2718             "line_chart"
2719           ]
2720         },
2721         "provider": {
2722           "type": "string",
2723           "enum": [
2724             "AWS",
2725             "AZURE",
2726             "GCP"
2727           ]
2728         },
2729         "accountId": {
2730           "type": "string",
2731           "format": "uuid"
2732         },
2733         "resourceId": {
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2734         "type": "string",
2735         "format": "uuid"
2736     },
2737     "metricType": {
2738         "type": "string"
2739     },
2740     "metricName": {
2741         "type": "string"
2742     }
2743 }
2744 }
2745 ]
2746 },
2747 "KpiWidgetConfigUpdateDTO": {
2748     "allOf": [
2749         {
2750             "$ref": "#/components/schemas/WidgetConfigUpdateDTO"
2751         },
2752         {
2753             "type": "object",
2754             "properties": {
2755                 "id": {
2756                     "type": "string",
2757                     "format": "uuid"
2758                 },
2759                 "widgetType": {
2760                     "type": "string",
2761                     "enum": [
2762                         "KPI",
2763                         "CHART"
2764                     ]
2765                 },
2766                 "displayName": {
2767                     "type": "string"
2768                 },
2769                 "chargeIds": {
2770                     "type": "array",
2771                     "items": {
2772                         "type": "string"
2773                     }
2774                 },
2775                 "aggregationWindowDays": {
2776                     "type": "integer",
2777                     "format": "int32"
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2778         }
2779     }
2780 }
2781 ]
2782 },
2783 "WidgetConfigUpdateDTO": {
2784     "discriminator": {
2785         "propertyName": "widgetType"
2786     },
2787     "properties": {
2788         "widgetType": {
2789             "type": "string"
2790         }
2791     },
2792     "required": [
2793         "widgetType"
2794     ]
2795 },
2796 "UpdateResourceStatusRequest": {
2797     "type": "object",
2798     "properties": {
2799         "status": {
2800             "type": "string",
2801             "enum": [
2802                 "active",
2803                 "disabled"
2804             ]
2805         }
2806     }
2807 },
2808 "UpdateAccountNameRequest": {
2809     "type": "object",
2810     "properties": {
2811         "name": {
2812             "type": "string"
2813         }
2814     }
2815 },
2816 "SseEmitter": {
2817     "type": "object",
2818     "properties": {
2819         "timeout": {
2820             "type": [
2821                 "integer",
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2822         "null"
2823     ],
2824     "format": "int64"
2825 }
2826 }
2827 },
2828 "TimestampedNumericDataPoint": {
2829     "type": "object",
2830     "properties": {
2831         "value": {
2832             "type": "number"
2833         },
2834         "timestamp": {
2835             "type": "string",
2836             "format": "date-time"
2837         }
2838     }
2839 },
2840 "BillingChargeResponse": {
2841     "type": "object",
2842     "properties": {
2843         "resourceId": {
2844             "type": "string"
2845         },
2846         "chargeId": {
2847             "type": "string"
2848         },
2849         "service": {
2850             "type": "string"
2851         },
2852         "provider": {
2853             "type": "string",
2854             "enum": [
2855                 "AWS",
2856                 "AZURE",
2857                 "GCP"
2858             ]
2859         }
2860     }
2861 },
2862 "CloudAccount": {
2863     "type": "object",
2864     "properties": {
2865         "id": {
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2866         "type": "string",
2867         "format": "uuid"
2868     },
2869     "connectionId": {
2870         "type": "string",
2871         "format": "uuid"
2872     },
2873     "connection": {
2874         "$ref": "#/components/schemas/CloudConnection"
2875     },
2876     "accountType": {
2877         "type": "string",
2878         "enum": [
2879             "aws_account",
2880             "azure_subscription",
2881             "gcp_project"
2882         ]
2883     },
2884     "displayName": {
2885         "type": "string"
2886     },
2887     "ingestionPeriod": {
2888         "type": "string"
2889     },
2890     "createdAt": {
2891         "type": "string",
2892         "format": "date-time"
2893     },
2894     "lastUsageIngestion": {
2895         "type": "string",
2896         "format": "date-time"
2897     },
2898     "nextUsageIngestion": {
2899         "type": "string",
2900         "format": "date-time"
2901     },
2902     "lastBillingIngestion": {
2903         "type": "string",
2904         "format": "date-time"
2905     },
2906     "nextBillingIngestion": {
2907         "type": "string",
2908         "format": "date-time"
2909     }
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2910     }
2911 },
2912 "CloudConnection": {
2913   "type": "object",
2914   "properties": {
2915     "id": {
2916       "type": "string",
2917       "format": "uuid"
2918     },
2919     "userId": {
2920       "type": "string",
2921       "format": "uuid"
2922     },
2923     "user": {
2924       "$ref": "#/components/schemas/User"
2925     },
2926     "provider": {
2927       "type": "string",
2928       "enum": [
2929         "AWS",
2930         "AZURE",
2931         "GCP"
2932       ]
2933     },
2934     "status": {
2935       "type": "string",
2936       "enum": [
2937         "active",
2938         "disabled"
2939       ]
2940     }
2941   }
2942 },
2943 "User": {
2944   "type": "object",
2945   "properties": {
2946     "id": {
2947       "type": "string",
2948       "format": "uuid"
2949     },
2950     "email": {
2951       "type": "string"
2952     },
2953     "username": {
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2954         "type": "string"
2955     },
2956     "passwordHash": {
2957         "type": "string"
2958     },
2959     "createdAt": {
2960         "type": "string",
2961         "format": "date-time"
2962     }
2963 },
2964 },
2965 "ResourceCountResponse": {
2966     "type": "object",
2967     "properties": {
2968         "count": {
2969             "type": "integer",
2970             "format": "int64"
2971         }
2972     }
2973 },
2974 "Resource": {
2975     "type": "object",
2976     "properties": {
2977         "id": {
2978             "type": "string",
2979             "format": "uuid"
2980         },
2981         "accountId": {
2982             "type": "string",
2983             "format": "uuid"
2984         },
2985         "account": {
2986             "$ref": "#/components/schemas/CloudAccount"
2987         },
2988         "resourceType": {
2989             "type": "string"
2990         },
2991         "resourceName": {
2992             "type": "string"
2993         },
2994         "resourceIdentifier": {
2995             "type": "string"
2996         },
2997         "resourceIdentifierType": {
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2998         "type": "string"
2999     },
3000     "region": {
3001         "type": "string"
3002     },
3003     "status": {
3004         "type": "string",
3005         "enum": [
3006             "active",
3007             "disabled"
3008         ]
3009     },
3010     "tags": {
3011         "type": "object",
3012         "additionalProperties": {}
3013     },
3014     "lastUpdated": {
3015         "type": "string",
3016         "format": "date-time"
3017     },
3018     "createdAt": {
3019         "type": "string",
3020         "format": "date-time"
3021     }
3022 },
3023 "CloudAccountDetailsResponse": {
3024     "type": "object",
3025     "properties": {
3026         "id": {
3027             "type": "string",
3028             "format": "uuid"
3029         },
3030         "displayName": {
3031             "type": "string"
3032         },
3033         "accountType": {
3034             "type": "string",
3035             "enum": [
3036                 "aws_account",
3037                 "azure_subscription",
3038                 "gcp_project"
3039             ]
3040         }
3041     },
```

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3042         "accountEmail": {
3043             "type": "string"
3044         },
3045         "ingestionPeriod": {
3046             "type": "string"
3047         },
3048         "createdAt": {
3049             "type": "string",
3050             "format": "date-time"
3051         }
3052     },
3053 },
3054 "ResourceNameDto": {
3055     "type": "object",
3056     "properties": {
3057         "resourceId": {
3058             "type": "string"
3059         },
3060         "resourceName": {
3061             "type": "string"
3062         }
3063     }
3064 },
3065 "ResourceMetric": {
3066     "type": "object",
3067     "properties": {
3068         "metricName": {
3069             "type": "string"
3070         },
3071         "metricType": {
3072             "type": "string"
3073         }
3074     }
3075 },
3076 "ResourceMetricsGroupDto": {
3077     "type": "object",
3078     "properties": {
3079         "resourceId": {
3080             "type": "string",
3081             "format": "uuid"
3082         },
3083         "metrics": {
3084             "type": "array",
3085             "items": {
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3086         "$ref": "#/components/schemas/ResourceMetric"
3087     }
3088 }
3089 }
3090 },
3091 "AggregatedMetric": {
3092     "type": "object",
3093     "properties": {
3094         "resourceId": {
3095             "type": "string",
3096             "format": "uuid"
3097         },
3098         "metricType": {
3099             "type": "string"
3100         },
3101         "metricName": {
3102             "type": "string"
3103         },
3104         "metricValue": {
3105             "type": "number"
3106         },
3107         "unit": {
3108             "type": "string"
3109         },
3110         "periodStart": {
3111             "type": "string",
3112             "format": "date-time"
3113         },
3114         "periodEnd": {
3115             "type": "string",
3116             "format": "date-time"
3117         },
3118         "sampleCount": {
3119             "type": "integer",
3120             "format": "int64"
3121         }
3122     }
3123 },
3124 "MetricDto": {
3125     "type": "object",
3126     "properties": {
3127         "resourceId": {
3128             "type": "string",
3129             "format": "uuid"
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3130     },
3131     "metricType": {
3132       "type": "string"
3133     },
3134     "metricName": {
3135       "type": "string"
3136     },
3137     "metricValue": {
3138       "type": "number"
3139     },
3140     "unit": {
3141       "type": "string"
3142     },
3143     "periodStart": {
3144       "type": "string",
3145       "format": "date-time"
3146     },
3147     "periodEnd": {
3148       "type": "string",
3149       "format": "date-time"
3150     },
3151     "sampleCount": {
3152       "type": "integer",
3153       "format": "int64"
3154     }
3155   }
3156 }
3157 }
3158 }
3159 }
```